Data Mining

Lecture Notes (Ch. 1–12)

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Chapter 1. Statistical Learning, Data and Exploration

Lectures 1–3 (overview of statistics/ML/AI, statistical learning, data as probabilistic evidence, assumptions, data structures, EDA)

1 Statistics, Machine Learning, AI and Data Mining

1.1 Where are we? A statistician's view

- **Statistics**: modeling the data-generating process under uncertainty.
- **Statistical learning**: estimating $P(Y \mid X)$ from limited samples.
- **Machine learning**: optimization for prediction.
- **Data mining**: discovery of patterns and structure in complex data.

AI $\supset$ ML $\supset$ deep learning; statistics underpins many ML methods and models uncertainty.

Example: one problem, four perspectives — customer churn

AI: decide retention strategies. **ML**: learn $\hat{f}(X)$ for the churn probability. **Statistics**: which variables matter, and how certain are we? **Data mining**: hidden segments and behavioral structure.

Data are *traces of a hidden generative process*; learning tries to understand that process.

2 What Is Statistical Learning?

Statistical learning estimates unknown probabilistic relationships from finite data.

- **Supervised learning**: estimate $P(Y \mid X)$ from labeled data.
- **Unsupervised learning**: uncover structure in $P(X)$ without labels.

Definition: notation

Outcome (response, target) Y; predictor vector $X = (X_1, \ldots, X_p)$; data $\{(X_i, Y_i)\}_{i=1}^n$. **Regression** if Y is quantitative, **classification** if Y is categorical.

Supervised learning: model $P(Y \mid X)$	Unsupervised learning: model $P(X)$
prediction: perform well on unseen data **inference**: understand how X influences Y **uncertainty**: reliability of predictions and estimates	**structure estimation**: properties of $P(X)$ **clustering**: latent groups, multimodality **representation learning**: low-dimensional summaries of variability

Learning is *generalization under uncertainty.*

2.1 Applications

Medical risk prediction, spam detection, handwritten-digit recognition, income versus demographics.

Example: spam data

4601 e-mails to "George" (HP Labs), labeled spam or e-mail; features: relative frequencies of 57 common words and punctuation marks. Largest spam/e-mail differences: `george` (0.00 vs 1.27%), `you` (2.26 vs 1.27), `hp` (0.02 vs 0.90), `free` (0.52 vs 0.07), `!` (0.51 vs 0.11), `edu` (0.01 vs 0.29), `remove` (0.28 vs 0.01).

3 Data as Probabilistic Evidence

3.1 Sample versus population

Observations are realizations of random variables from an unknown data-generating process: **stochastic evidence, not absolute truth**.

Definition: the inferential setting

Observed sample $\mathcal{D}_n = \{Z_1, \dots, Z_n\}$, population distribution $Z \sim P_0$, target $\theta(P_0)$ (a center, a dispersion, a regression coefficient, ...).

Sampling variability. Repeating n coin tosses gives a different $\hat{p} = \bar{Z}$ each time, $\mathrm{Var}(\hat{p}) = p(1-p)/n$.

3.2 Assumptions and identifiability

Data alone do not determine the target, $\mathcal{D}_n \not\Rightarrow \theta(P_0)$; we restrict $P_0 \in \mathcal{P}_{\text{model}}$, within which identification becomes possible.

Example: five rainy days

Rain, Rain, Rain, Rain, Rain. Does it always rain, did the rainy season start, or were we unlucky? Only an assumption (i.i.d. days, a seasonal model) picks one explanation.

Definition: data-generating process

$$Y = f(X) + \varepsilon,$$

where f is the underlying mechanism and ε captures randomness and unobserved factors.

A model specifies a **class** for f, which fixes flexibility and tools:

Function class	Example	Tools
linear models	$f(x) = x^\top \beta$	linear algebra, projections, least squares
parametric models	$f(x) = f(x;\theta)$, $\theta \in \Theta$	likelihood theory, asymptotics, Fisher information
RKHS (kernel) models	$f \in \mathcal{H}_K$	Hilbert spaces, kernels, representer theorem
nonparametric classes	Sobolev, Hölder balls	functional analysis, approximation theory

4 Data Structures and Their Implications

4.1 The i.i.d. assumption

$Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} P_0$ (randomized trials, survey responses, coin tosses) gives the LLN, the CLT, $\sqrt{n}$-convergence, and classical confidence intervals, tests and regression inference.

4.2 Beyond i.i.d.

- **Dependence**: with positive autocorrelation the effective sample size is below n; for a stationary AR(1), $\mathrm{Var}(\bar{Z}) \approx \frac{\sigma^2}{n}\frac{1+\rho}{1-\rho}$.
- **High dimension**: $p \gg n$. When $p > n$, $Y = X\beta$ is underdetermined and OLS is not unique; structure such as sparsity $\|\beta\|_0 = s \ll n$ is needed.
- **Curse of dimensionality**: covering $[0,1]^p$ by cubes of side ϵ needs $N(\epsilon) \asymp (1/\epsilon)^p$ cubes, so local methods (Ch. 2) need exponentially many observations.
- **Partial observation**: with survival time T and censoring time C we record $\tilde{T} = \min(T, C)$.
- **Measurement error**: $\tilde{X} = X + \delta$ and $Y = f(X) + \varepsilon$; observed relationships may be distorted (attenuation, Ch. 5).

Derivation: why the curse of dimensionality bites local methods

Let $X \sim \mathrm{Unif}[0,1]^p$. A cube capturing a fraction r of the data has side $e_p(r) = r^{1/p}$; for $r = 0.01$, $p = 10$, $e_p \approx 0.63$, so the neighborhood is not local. For $n = 500$ uniform points in the unit ball with $p = 10$, the median distance from the center to the nearest point is about 0.52, tending to 1 as p grows (Exercise 1.2).

5 Exploratory Data Analysis

5.1 Visualization

Distributions of each variable, relationships, correlation and clusters, projections, temporal and spatial patterns, anomalies, missing data, model diagnostics.

5.2 The geometry of the data cloud

Gaussian density contours are ellipses determined by the covariance matrix.

Aspect	Summary	Classical test
center (location)	mean	z-test, t-test
dispersion (spread)	standard deviation	χ^2-test, F-test
orientation (direction)	correlation	t-test for correlation

Shape: skewness $\mathbb{E}[(X-\mu)^3]/\sigma^3$ and kurtosis $\mathbb{E}[(X-\mu)^4]/\sigma^4$ (3 for the normal).

5.3 Other summaries

Quantiles, the five-number summary $(\min, Q_1, \text{median}, Q_3, \max)$, robust statistics (median, IQR $= Q_3 - Q_1$, MAD $= \text{median}\,|X_i - \text{median}\,X|$), boxplots and density plots. One wild value can move the mean arbitrarily far but not the median beyond the range of the remaining data.

Summary

- Statistical learning estimates $P(Y \mid X)$ or the structure of $P(X)$ from finite, noisy samples.
- Without assumptions the target is not identified; $Y = f(X) + \varepsilon$ with a function class fixes flexibility and tools.
- Dependence, high dimension, censoring and measurement error change what can be learned.

6 Selected exercises

Motivation. Short derivations behind four claims of this chapter.

Exercise 1.1 (Variance of the mean under AR(1) dependence)

Let (Z_t) be stationary with mean μ, variance $\sigma^2 > 0$ and $\text{Corr}(Z_s, Z_t) = \rho^{|s-t|}$, $|\rho| < 1$.
(a) Show that $\text{Var}(\bar{Z}_n) = \frac{\sigma^2}{n}\Big[1 + 2\sum_{h=1}^{n-1}\Big(1 - \frac{h}{n}\Big)\rho^h\Big]$.
(b) Show that $n\text{Var}(\bar{Z}_n) \to \sigma^2 \frac{1+\rho}{1-\rho}$.
(c) Interpret $n_{\text{eff}} = n(1-\rho)/(1+\rho)$.

Solution [Supplement].

(a) $\text{Var}(\sum_t Z_t) = \sigma^2 \sum_{s,t} \rho^{|s-t|}$; n pairs have $|s-t| = 0$ and $2(n-h)$ have $|s-t| = h$, $1 \le h \le n-1$. Hence

$$\text{Var}\Big(\sum_t Z_t\Big) = \sigma^2\Big[n + 2\sum_{h=1}^{n-1}(n-h)\rho^h\Big];$$

divide by n^2.

(b) From (a),

$$n\text{Var}(\bar{Z}_n) = \sigma^2\Big[1 + 2\sum_{h=1}^{n-1}\rho^h - \frac{2}{n}\sum_{h=1}^{n-1} h\rho^h\Big].$$

The first sum tends to $\rho/(1-\rho)$; since $\sum_h h|\rho|^h < \infty$, the last term is $\le \frac{2}{n}\sum_{h\ge 1} h|\rho|^h \to 0$. The limit is $\sigma^2\{1 + 2\rho/(1-\rho)\} = \sigma^2(1+\rho)/(1-\rho)$.

(c) $\text{Var}(\bar{Z}_n) \approx \sigma^2/n_{\text{eff}}$: the sample is as informative as n_{eff} independent observations. $\rho = 0.5$ gives $n_{\text{eff}} = n/3$; for $\rho < 0$ the mean is more precise than under independence. ■

Exercise 1.2 (The nearest neighbor is far away)

Let $X_1, \ldots, X_n$ be i.i.d. uniform on $B_p = \{x \in \mathbb{R}^p : \|x\| \le 1\}$ and $D = \min_i \|X_i\|$.
(a) Show that the median of D is $d(p,n) = \big(1 - 2^{-1/n}\big)^{1/p}$.
(b) Evaluate $d(10, 500)$ and show that $d(p,n) \to 1$ as $p \to \infty$ for fixed n.

Solution [Supplement].

(a)

Step 1. $\mathrm{vol}(rB_p) = r^p \,\mathrm{vol}(B_p)$, so $\Pr(\|X_1\| \le r) = r^p$, $0 \le r \le 1$.

Step 2. $\Pr(D > r) = \prod_i \Pr(\|X_i\| > r) = (1 - r^p)^n$, continuous and strictly decreasing from 1 to 0 on $[0, 1]$. The median solves $(1 - m^p)^n = 1/2$, i.e. $m^p = 1 - 2^{-1/n}$.

(b) $d(10, 500) \approx 0.518$. For fixed n, $c = 1 - 2^{-1/n} \in (0, 1)$ and $c^{1/p} = e^{(\log c)/p} \to 1$: in high dimension the "nearest" neighbor is not near. ■

Exercise 1.3 (The median cannot be carried away)

Let $n = 2m + 1$ and $x_1, \ldots, x_n \in \mathbb{R}$. Replace any $k \le m$ observations by arbitrary values; let U be the $n - k$ untouched values, $a = \min U$, $b = \max U$.
(a) Show that the median of the new sample lies in $[a, b]$.
(b) Show that with $m + 1$ replaced values the median can be made arbitrarily large, whereas one replaced value suffices for the mean. Conclude that the breakdown points are $(n + 1)/(2n) \to 1/2$ for the median and $1/n$ for the mean.

Solution [Supplement].

(a) Let $y_{(1)} \le \cdots \le y_{(n)}$ be the new sample, median $y_{(m+1)}$. At most $k \le m$ values are $< a$; if $y_{(m+1)} < a$, the $m + 1$ values $y_{(1)}, \ldots, y_{(m+1)}$ would all be $< a$. Likewise at most m values are $> b$; if $y_{(m+1)} > b$, the $m + 1$ values $y_{(m+1)}, \ldots, y_{(n)}$ would all be $> b$.

(b) Set $m + 1$ values equal to $T > \max_i x_i$; then $y_{(m+1)} = T$, arbitrarily large. Replacing x_1 by $x_1 + t$ moves $\bar{x}$ by t/n, unbounded in t. By (a) no fraction $k/n \le m/n$ breaks the median, so its breakdown point is $(m + 1)/n = (n + 1)/(2n)$. ■

Exercise 1.4 (Least squares when $p > n$)

Let $X \in \mathbb{R}^{n \times p}$ with $p > n$ and $y \in \mathbb{R}^n$.
(a) Show that $X^\top X$ is singular and that the normal equations $X^\top X\beta = X^\top y$ have infinitely many solutions.
(b) Show that all solutions give the same fitted vector $X\hat{\beta}$; if $\mathrm{rank} X = n$, this vector is y itself (zero training error).
(c) If $\mathrm{rank} X = n$, show that $\beta^\dagger = X^\top (XX^\top)^{-1} y$ is the unique solution of minimum Euclidean norm.

Solution [Supplement].

(a) $\mathrm{rank}(X^\top X) = \mathrm{rank} X \le n < p$, so $X^\top X$ is singular and $\dim \mathcal{N}(X) \ge p - n \ge 1$. Since $\mathrm{col}(X^\top X) \subseteq \mathrm{col}(X^\top)$ with equal dimensions, $X^\top y \in \mathrm{col}(X^\top X)$: the system is consistent. If $\hat{\beta}$ solves it, so does $\hat{\beta} + v$ for every $v \in \mathcal{N}(X)$.

(b) For solutions β_1, β_2, $\|X(\beta_1 - \beta_2)\|^2 = (\beta_1 - \beta_2)^\top X^\top X(\beta_1 - \beta_2) = 0$, so $X\beta_1 = X\beta_2$. If $\mathrm{rank} X = n$, $\mathrm{col}(X) = \mathbb{R}^n$, so some β has $X\beta = y$; it solves the normal equations, hence every solution has fitted vector y.

(c) $XX^\top$ is invertible and $X\beta^\dagger = y$; the solution set of $X\beta = y$ is $\beta^\dagger + \mathcal{N}(X)$. As $\beta^\dagger = X^\top w \in \mathcal{N}(X)^\perp$, $\|\beta^\dagger + v\|^2 = \|\beta^\dagger\|^2 + \|v\|^2$ for $v \in \mathcal{N}(X)$, minimized only at $v = 0$. ■

Chapter 2. Supervised Learning and Model Assessment

Lecture 4 (regression function, parametric vs nonparametric, KNN, bias–variance, adjusted R^2, AIC, BIC)

1 The Supervised Learning Setup

Motivation. $Y = f(X) + \epsilon$ separates what can be learned (f) from what cannot (ϵ), so even a perfect method has nonzero test error.

Example: advertising data

`Sales` and advertising budgets `TV`, `Radio`, `Digital` in many markets: $Y =$ `Sales`, $X = (X_1, X_2, X_3)$,

$$Y = f(X) + \varepsilon,$$

f unknown, $\mathbb{E}\varepsilon = 0$, ε independent of X. Similarly: income of 30 individuals versus education, seniority and marital status.

1.1 Why estimate f?

Prediction: $\hat{f}(x)$ at a new $X = x$. **Inference**: which predictors matter, and the direction and size of their effects (e.g. the change in sales from an extra \$1,000 on TV).

1.2 The regression function

At $X = 4$ many Y values are possible; the best prediction is:

Theorem: the conditional mean is the optimal predictor

$$f(x) = \arg\min_g \mathbb{E}[(Y - g(X))^2 \mid X = x] = \mathbb{E}(Y \mid X = x),$$

called the **regression function**. Moreover, for any g, $\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - f(X))^2] + \mathbb{E}[(f(X) - g(X))^2]$.

Proof [Supplement]. For a constant c, $\mathbb{E}[(Y - c)^2 \mid X = x] = \mathrm{Var}(Y \mid X = x) + (f(x) - c)^2$, minimized at $c = f(x)$. For the decomposition, write $Y - g = (Y - f) + (f - g)$; the cross term vanishes because $\mathbb{E}[(Y - f(X)) \mid X] = 0$. ■

$\mathrm{Var}(Y \mid X = x) = \mathrm{Var}(\varepsilon)$ is the **irreducible error**; $\mathbb{E}(f - g)^2$ is the **reducible error**.

2 Parametric Approach

1. specify a functional form, e.g. the linear model $f_L(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$;
2. estimate the parameters $(\beta_0, \ldots, \beta_p)$ from data (e.g. by least squares).

A low-dimensional parameter space gains interpretability and stability but risks **model misspecification**. More parameters, more flexibility: $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ versus $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$.

3 Nonparametric Approach

3.1 Local averaging

Few observations have exactly $X = 4$, so we average nearby points:

$$\hat{f}(x) = \text{Ave}(Y \mid X \in \mathcal{N}(x)),$$

with $\mathcal{N}(x)$ a neighborhood of x.

Definition: K-nearest neighbors regression

$$\hat{f}(x) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} y_i,$$

where $\mathcal{N}_K(x)$ is the set of the K training points closest to x.

Others: kernel regression, smoothing splines (Ch. 7), Gaussian processes. $K = 1$ interpolates the data; $K = 9$ is smoother.

Theorem: bias and variance of KNN at a point

Let $Y = f(X) + \varepsilon$ with $\text{Var}\varepsilon = \sigma^2$ and condition on the inputs. With neighbors $x_{(1)}, \ldots, x_{(K)}$ of x_0,

$$\mathbb{E}\big[(Y_0 - \hat{f}(x_0))^2\big] = \sigma^2 + \Big[f(x_0) - \frac{1}{K}\sum_{l=1}^{K} f(x_{(l)})\Big]^2 + \frac{\sigma^2}{K}.$$

Proof [Supplement]. $\hat{f}(x_0) = \frac{1}{K}\sum_l f(x_{(l)}) + \frac{1}{K}\sum_l \varepsilon_{(l)}$; its variance is σ^2/K and its mean is the average of f over the neighbors. Apply the bias–variance decomposition (§4.3). ■

Small K: low bias, high variance σ^2/K; large K: higher bias. *Smaller K = more* flexible.

3.2 Pros and cons

No assumption on the form of f, but a very large n is needed (curse of dimensionality, Ch. 1), and a very flexible fit can reproduce the training data and still predict badly.

4 Assessing Model Accuracy

4.1 Complexity, underfitting, overfitting

Underfitting: too simple a model, systematic errors. **Overfitting**: the model fits noise rather than signal.

4.2 Training error and test error

For a model fitted on $\text{Tr} = \{(x_i, y_i)\}_{i=1}^n$,

$$\text{MSE}_{\text{Tr}} = \text{Ave}_{i \in \text{Tr}}\big[y_i - \hat{f}(x_i)\big]^2, \qquad \text{MSE}_{\text{Te}} = \text{Ave}_{i \in \text{Te}}\big[y_i - \hat{f}(x_i)\big]^2,$$

with $\text{Te} = \{(x_i, y_i)\}_{i=1}^M$ fresh observations. Training error is optimistic, especially for flexible models. Simulations:

(1) moderately curved truth: training MSE falls monotonically, test MSE is U-shaped;

(2) nearly linear truth: the linear model and a smooth fit do best;

(3) wiggly truth, low noise: very flexible fits do best.

4.3 Bias–variance trade-off

Theorem: decomposition of the expected test error

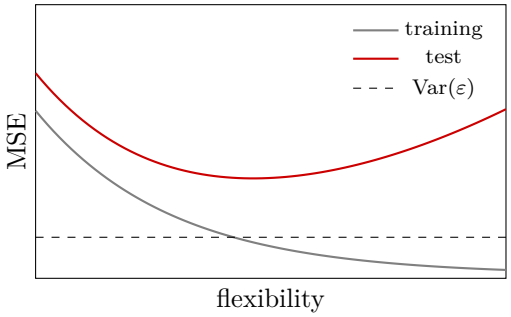
Let (x_0, y_0) be a new observation, $y_0 = f(x_0) + \varepsilon_0$, $f(x) = \mathbb{E}(Y \mid X = x)$, and let $\hat{f}$ be fitted on an independent training set. Then

$$\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{[\mathrm{Bias}(\hat{f}(x_0))]^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{irreducible}} .$$

Proof [Supplement]. $y_0 - \hat{f} = \varepsilon_0 + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$; the three terms are uncorrelated (ε_0 is independent of the training data and has mean 0; the second term is constant; the third has mean 0). Square and take expectations. ■

More flexibility: lower bias, higher variance.

(a) training vs. test error



(b) decomposition of test MSE

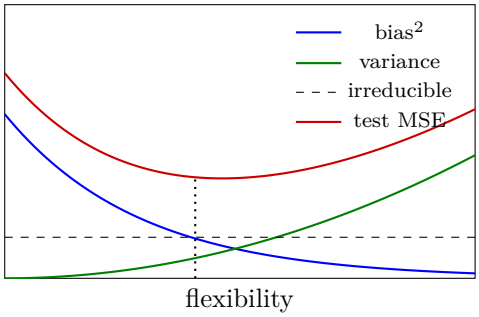
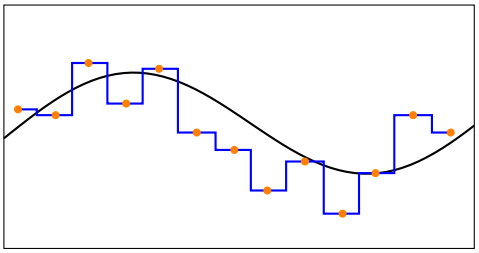


Figure 1: Bias–variance trade-off (schematic); dotted: optimal flexibility.

$K = 1$: low bias, high variance



$K = 9$: smoother, more bias

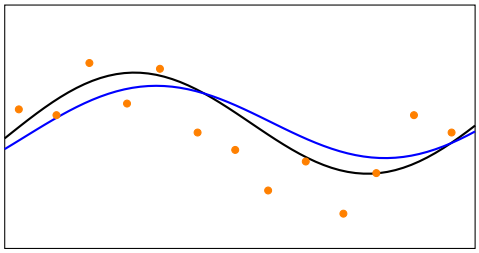


Figure 2: KNN regression (schematic). Black: true f; blue: fit.

5 Criteria Based on the Training Data

$$\text{Score} = \text{goodness of fit} + \text{complexity penalty}, \qquad \mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

5.1 Adjusted R^2

$R^2 = 1 - \text{RSS}/\text{TSS}$ always increases when a predictor is added;

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}.$$

It rises iff $\text{RSS}/(n-p-1)$ falls, i.e. iff the new predictor has $|t| > 1$ (Exercise 2.2).

5.2 AIC and BIC

For a Gaussian linear model with p predictors,

$$\text{AIC} = n\log(\text{RSS}/n) + 2(p+1), \qquad \text{BIC} = n\log(\text{RSS}/n) + \log(n)(p+1);$$

smallest value wins. In general, with k parameters, $\text{AIC} = -2\log L(\hat\theta) + 2k$, $\text{BIC} = -2\log L(\hat\theta) + k\log n$.

Derivation: information-theoretic interpretation of AIC

$\text{KL}(p_0 \| p_\theta) = \mathbb{E}_{p_0} \log \frac{p_0(X)}{p_\theta(X)}$ equals, up to a constant, $-\mathbb{E}_{p_0} \log p_\theta(X)$, the expected log-loss on a *new* observation; $\log L(\hat\theta)$ is optimistic since $\hat\theta$ was fitted to the same data. A second-order Taylor expansion around θ^* (correct model, regular case): fitting and evaluating on new data each contribute about $k/2$, so

$$\mathbb{E}\big[-2\,\mathbb{E}_{\text{new}} \log p_{\hat\theta}(X^{\text{new}})\big] \approx \mathbb{E}\big[-2\log L(\hat\theta)\big] + 2k.$$

so AIC estimates the expected out-of-sample deviance (Akaike).

Derivation: Bayesian interpretation of BIC

With equal prior weights, $P(M_k \mid \text{data}) \propto \int L_k(\theta)\pi_k(\theta)d\theta$; a Laplace approximation around $\hat\theta_k$ gives

$$\log P(M_k \mid \text{data}) \approx \log L(\hat\theta_k) - \frac{k}{2}\log n + O(1)$$

(Schwarz, 1978); times -2 this is BIC, so the smallest BIC has the largest posterior probability.

	Penalty	Goal	Model size	Consistent?
AIC	$2k$	prediction	larger models	no
BIC	$k\log n$	true-model selection	smaller models	yes

BIC penalizes more once $n \geq 8$ ($\log n > 2$). **Consistency**: if the true model is a candidate, BIC selects it with probability $\to 1$; AIC keeps a positive probability of overfitting.

Summary

- $\mathbb{E}(Y \mid X = x)$ is optimal under squared loss; $\text{Var}(\varepsilon)$ is irreducible.
- Parametric: interpretable, may be misspecified; KNN-type local averaging: flexible, data-hungry.
- Test error is U-shaped in flexibility (bias–variance); AIC targets prediction, BIC consistency.

6 Selected exercises

Motivation. Three derivations the chapter uses without proof.

Exercise 2.1 (Absolute loss and the conditional median)

(a) Let Y be a random variable with $\mathbb{E}|Y| < \infty$ and let m be a median, $\Pr(Y \le m) \ge \frac{1}{2}$ and $\Pr(Y \ge m) \ge \frac{1}{2}$. Show that $\mathbb{E}|Y - c| \ge \mathbb{E}|Y - m|$ for every $c \in \mathbb{R}$.
(b) Deduce that $\mathbb{E}|Y - g(X)|$ is minimized over functions g by a conditional median $g(x) = \text{median}(Y \mid X = x)$.

Solution [Supplement].

(a) All expectations below are finite because $\mathbb{E}|Y| < \infty$.

Step 1 ($c > m$). If $y \le m$, then $|y - c| - |y - m| = (c - y) - (m - y) = c - m$. For every y, the triangle inequality gives $|y - c| - |y - m| \ge -(c - m)$. Hence

$$\mathbb{E}\big[|Y - c| - |Y - m|\big] \ge (c - m)\Pr(Y \le m) - (c - m)\Pr(Y > m) = (c - m)\{2\Pr(Y \le m) - 1\} \ge 0.$$

Step 2 ($c < m$). Symmetrically, $|y - c| - |y - m| = m - c$ for $y \ge m$ and $\ge -(m - c)$ always, so the difference has expectation $\ge (m - c)\{2\Pr(Y \ge m) - 1\} \ge 0$.

(b) Let $m(x) = \inf\{c : \Pr(Y \le c \mid X = x) \ge \frac{1}{2}\}$ (a measurable conditional median). By (a) applied conditionally, $\mathbb{E}\big[|Y - g(X)| \,\big|\, X = x\big] \ge \mathbb{E}\big[|Y - m(X)| \,\big|\, X = x\big]$ for every x; take expectations over X. ■

Exercise 2.2 (Adjusted R^2 rises iff $|t| > 1$)

Let M_0 have design $X_0 \in \mathbb{R}^{n \times (p+1)}$ (intercept and p predictors) and residual sum of squares RSS_0, and let M_1 add one predictor z, with design $X_1 = [X_0, z]$ of full column rank $p + 2 < n$ and $\text{RSS}_1 > 0$. Let $t = \hat\gamma / \text{SE}(\hat\gamma)$ be the usual t-statistic of the coefficient $\hat\gamma$ of z in M_1, with $\hat\sigma^2 = \text{RSS}_1/(n - p - 2)$.
(a) Show that $\text{RSS}_0 - \text{RSS}_1 = t^2\,\text{RSS}_1/(n - p - 2)$.
(b) Show that $R^2_{\text{adj}}(M_1) > R^2_{\text{adj}}(M_0)$ if and only if $|t| > 1$.

Solution [Supplement].

(a) Let H_0 be the orthogonal projection onto $\text{col}(X_0)$ and $\tilde z = (I - H_0)z$; $\tilde z \neq 0$ since X_1 has full column rank.

Step 1. $\text{col}(X_1) = \text{col}(X_0) \oplus \text{span}(\tilde z)$ with $\tilde z \perp \text{col}(X_0)$, so the projection onto $\text{col}(X_1)$ is $H_1 = H_0 + \tilde z \tilde z^\top / (\tilde z^\top \tilde z)$.

Step 2. Since $\tilde z^\top H_0 = 0$, we have $\tilde z^\top y = \tilde z^\top (I - H_0) y$ and

$$\text{RSS}_1 = y^\top (I - H_1) y = \text{RSS}_0 - \frac{(\tilde z^\top y)^2}{\tilde z^\top \tilde z}.$$

Step 3. Put $c = \tilde z^\top y / \tilde z^\top \tilde z$. Then $H_1 y = H_0 y + c\tilde z = (H_0 y - cH_0 z) + cz$, where $H_0 y - cH_0 z \in \text{col}(X_0)$. The coefficients of $H_1 y$ in the basis given by the columns of X_1 are unique, so $\hat\gamma = c$. As a linear function of y, $\text{Var}(\hat\gamma) = \sigma^2 / \tilde z^\top \tilde z$, so $\text{SE}(\hat\gamma)^2 = \hat\sigma^2 / \tilde z^\top \tilde z$.

Step 4. Hence $t^2 = \dfrac{(\tilde z^\top y)^2}{\tilde z^\top \tilde z\, \hat\sigma^2} = \dfrac{\text{RSS}_0 - \text{RSS}_1}{\hat\sigma^2}$, which is (a).

(b) Both models share TSS, so M_1 has the larger R^2_{adj} iff $\text{RSS}_1/d < \text{RSS}_0/(d+1)$ with $d = n - p - 2$. By (a), $\text{RSS}_0 = \text{RSS}_1(1 + t^2/d)$, so the condition reads $d + 1 < d + t^2$, i.e. $t^2 > 1$. ■

Exercise 2.3 (The Gaussian log-likelihood in AIC and BIC)

Let $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n\times(p+1)}$ of full column rank, and assume $\mathrm{RSS} > 0$ for the least-squares fit.
(a) Show that $\max_{\beta,\sigma^2} \log L(\beta, \sigma^2) = -\frac{n}{2}\{\log(2\pi) + \log(\mathrm{RSS}/n) + 1\}$.
(b) Deduce that counting σ^2 as a parameter ($k = p + 2$) changes AIC and BIC only by a constant common to all models fitted to the same n observations, so the model rankings agree with the formulas of this chapter.

Solution [Supplement].

(a)

Step 1. $\log L = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - X\beta\|^2$. For each fixed σ^2 this is maximized by minimizing $\|y - X\beta\|^2$, i.e. at the least-squares $\hat\beta$, giving $h(s) = -\frac{n}{2}\log(2\pi s) - \mathrm{RSS}/(2s)$ with $s = \sigma^2$.

Step 2. $h'(s) = -\frac{n}{2s} + \frac{\mathrm{RSS}}{2s^2} = \frac{\mathrm{RSS} - ns}{2s^2}$ is positive for $s < \mathrm{RSS}/n$ and negative beyond, so h is maximized at $\hat\sigma^2 = \mathrm{RSS}/n$, where $h = -\frac{n}{2}\log(2\pi\,\mathrm{RSS}/n) - \frac{n}{2}$.

(b) By (a), $-2\log L(\hat\theta) = n\log(\mathrm{RSS}/n) + n(1 + \log 2\pi)$. With $k = p + 2$,

$$\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2(p+1) + \{n(1+\log 2\pi) + 2\},$$
$$\mathrm{BIC} = n\log(\mathrm{RSS}/n) + (p+1)\log n + \{n(1+\log 2\pi) + \log n\}.$$

The braces do not depend on the model, so differences between models, and hence the selected model, are unchanged. ■

Chapter 3. Resampling: Cross-Validation and the Bootstrap

Lecture 5 (validation set, LOOCV, K-fold CV, right and wrong CV, bootstrap SE and CI)

1 What Are Resampling Methods?

Refit the model on repeated samples from the training set for

- **model assessment**: estimate test error rates;
- **model selection**: choose the level of flexibility.

Without a large test set, AIC/BIC adjust the training error (Ch. 2); here we hold out observations instead.

2 Cross-Validation

2.1 Validation-set approach

Randomly split into training and validation sets; the validation error (MSE, or misclassification rate) estimates the test error.

Limitations. (i) Highly variable, depending on the split.

(ii) Fitting on part of the data *overestimates* the test error of the full-data fit.

2.2 Leave-one-out cross-validation

Hold out one observation, fit on $n-1$, compute $\mathrm{MSE}_i = (y_i - \hat{y}_{(i)})^2$; repeat n times:

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i.$$

Less bias than the validation set, no split randomness, but n fits.

Theorem: LOOCV shortcut for least squares and linear smoothers

If $\hat{y} = Hy$ for a fixed matrix H with the leave-one-out property (true for least squares, ridge, smoothing splines), then

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

so LOOCV costs a single fit.

Proof for least squares [Supplement]. By the Sherman–Morrison formula, removing row i changes the estimate by $\hat{\beta} - \hat{\beta}_{(i)} = (X^\top X)^{-1} x_i e_i/(1-h_{ii})$ with $e_i = y_i - \hat{y}_i$. Hence $y_i - x_i^\top \hat{\beta}_{(i)} = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$. ■

High-leverage residuals are inflated.

2.3 K-fold cross-validation

Divide the data into K roughly equal parts $C_1, \ldots, C_K$ of sizes n_k; fit without part k and predict it:

$$\mathrm{CV}_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k, \qquad \mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2,$$

$\hat{y}_i$ fitted without part k; $K = n$ gives LOOCV.

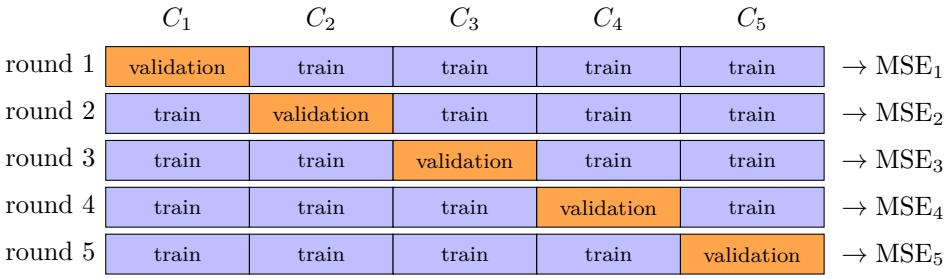


Figure 3: 5-fold cross-validation; each part is held out once.

2.4 K-fold CV versus LOOCV

Typically $K = 5$ or 10. Besides computation: the n LOOCV fits use nearly identical data, so their errors are *highly correlated* and the average has high variance. K-fold CV has more bias (training size $(K-1)n/K$) but lower variance. On the Auto data, LOOCV and nine repetitions of 10-fold CV give very similar curves.

2.5 Cross-validation: right and wrong

Example: selecting predictors before CV

5000 predictors, 200 samples:
(1) select the 100 predictors most correlated with the response;
(2) fit a model on these 100. CV on step 2 only? **No**: step 1 used all responses, so even for pure-noise predictors the CV error is near 0 while the true error is 50%. Right: in each fold, do steps 1 and 2 on the training part only.

Feature selection is training and must be inside the CV loop.

2.6 What does CV estimate?

CV estimates the prediction risk $R(\hat{f}) = \mathbb{E}[(Y - \hat{f}(X))^2]$.

Theory and limitation. LOOCV is asymptotically equivalent to AIC (Stone, 1977), but may prefer overly complex models, i.e. is not consistent for selection (Shao, 1997). Prediction $\rightarrow$ CV, AIC; true-model identification $\rightarrow$ BIC.

3 The Bootstrap

A tool to quantify the uncertainty of an estimator, e.g. a standard error or a confidence interval.

3.1 A motivating example

Invest α in an asset with return X and $1-\alpha$ in one with return Y; minimize $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem: minimum-variance allocation

With $\sigma_X^2 = \mathrm{Var}X$, $\sigma_Y^2 = \mathrm{Var}Y$, $\sigma_{XY} = \mathrm{Cov}(X,Y)$,

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof [Supplement]. $V(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$ is a convex quadratic; $V'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0$ gives $\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}$. ■

Plug in estimates:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

Simulation (true $\alpha = 0.6$, 100 pairs): four data sets gave $\hat{\alpha} = 0.576, 0.532, 0.657, 0.651$; 1,000 repetitions gave $\bar{\alpha} = 0.5996$ and $\sqrt{\frac{1}{999}\sum_r(\hat{\alpha}_r - \bar{\alpha})^2} = 0.083 \approx \mathrm{SE}(\hat{\alpha})$.

3.2 Back to the real world

New population samples are unavailable; the bootstrap samples n observations **from the original data with replacement**. Histograms of 1,000 true-resampling and 1,000 bootstrap estimates look very similar.

Algorithm: bootstrap standard error

For $b = 1, \ldots, B$ (say $B = 100$ or 1000): draw Z^{*b} by sampling n observations with replacement, and compute $\hat{\alpha}^{*b}$. Then

$$\mathrm{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}\right)^2}$$

estimates the standard error of $\hat{\alpha}$ computed from the original data.

Why it works. *Plug-in*: replace P_0 by $\hat{P}_n$ (mass $1/n$ per observation); for smooth statistics the law of $\hat{\alpha}^* - \hat{\alpha}$ approximates that of $\hat{\alpha} - \alpha$.

3.3 Bootstrap in general

For time series, the **block bootstrap** resamples blocks of consecutive observations, preserving short-range dependence.

3.4 Confidence intervals

The 5% and 95% quantiles of 1,000 bootstrap $\hat{\alpha}^*$ are $(0.43, 0.72)$: an approximate 90% **bootstrap percentile interval**. Alternatives: normal $\hat{\alpha} \pm z_{1-\alpha/2}\mathrm{SE}_B$; basic $(2\hat{\alpha} - q_{0.95},\ 2\hat{\alpha} - q_{0.05})$.

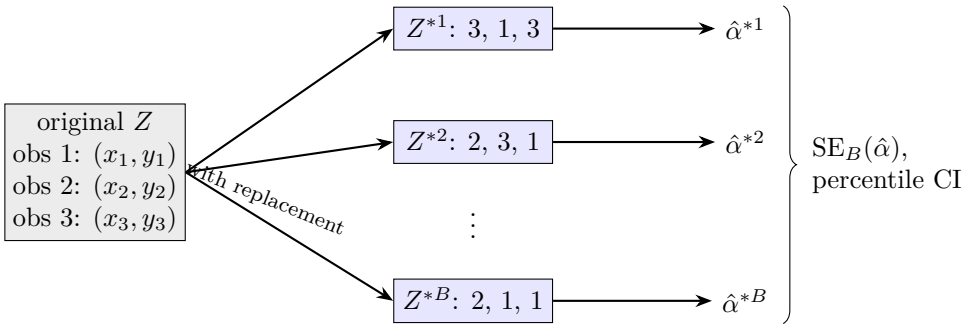


Figure 4: The bootstrap with $n = 3$.

3.5 The wrong way: bootstrap prediction error

Training on bootstrap samples and validating on the original sample: an observation is in a bootstrap sample with probability $1 - (1 - 1/n)^n \approx 0.632$ (Exercise 3.1), so the prediction error is seriously *underestimated*. Out-of-bag observations fix this (".632" estimators, OOB error in bagging, Ch. 11).

Summary

- Validation set: variable, biased upward; LOOCV: nearly unbiased, highly correlated fold errors; $K = 5$ or 10 balances both.
- Every data-dependent step (including variable selection) must be inside the CV loop.
- Prediction → cross-validation (like AIC); uncertainty → bootstrap.

4 Selected exercises

Motivation. The 0.632 inclusion probability (ISLR Ex. 5.2), the ridge LOOCV shortcut, and the exact bootstrap SE of a mean.

Exercise 3.1 (Inclusion in a bootstrap sample; ISLR Ex. 5.2)

A bootstrap sample draws n observations with replacement from n observations.
(a) What is the probability that the first bootstrap observation is not the jth original observation? The second?
(b) Show that the probability that observation j is not in the bootstrap sample is $(1-1/n)^n$, and find the expected number of distinct original observations in a bootstrap sample.
(c) Evaluate the inclusion probability for $n = 5, 100, 10\,000$, and prove that it decreases strictly to $1 - e^{-1}$ as $n \to \infty$ ($n \geq 2$).

Solution.

(a) Each draw is uniform on $\{1, \ldots, n\}$, so the first draw differs from j with probability $1 - 1/n$; the same holds for the second draw.

(b) The n draws are independent, so $\Pr(j \notin \text{sample}) = (1 - 1/n)^n$. With I_j the indicator that j appears, the number of distinct observations is $\sum_j I_j$, with expectation $n\{1 - (1 - 1/n)^n\}$ by linearity.

(c) The values $1 - (1 - 1/n)^n$ are 0.672, 0.634 and 0.632.

Step 1. Write $(1-1/n)^n = e^{a_n}$ with $a_n = \varphi(1/n)$, where $\varphi(x) = \log(1-x)/x$ for $0 < x < 1$. From $\log(1-x) = -\sum_{k\geq 1} x^k/k$,

$$\varphi(x) = -\Big(1 + \frac{x}{2} + \frac{x^2}{3} + \cdots\Big),$$

which is strictly decreasing in x with $\varphi(x) \to -1$ as $x \downarrow 0$.

Step 2. As n increases, $x = 1/n$ decreases, so $a_n \uparrow -1$ strictly and $(1-1/n)^n \uparrow e^{-1}$ strictly; the inclusion probability decreases strictly to $1 - e^{-1} \approx 0.632$. ■

Exercise 3.2 (LOOCV shortcut for ridge regression)

Let $X \in \mathbb{R}^{n\times p}$ (rows $x_i^\top$, no intercept), $\lambda > 0$, $A = X^\top X + \lambda I$, $\hat\beta = A^{-1}X^\top y$, $\hat y = X\hat\beta$, and $h_{ii} = x_i^\top A^{-1} x_i$. Let $\hat\beta_{(i)}$ be the ridge estimate with the same λ computed without observation i.
(a) Prove the Sherman–Morrison formula: if A is invertible and $c = 1 - u^\top A^{-1}u \neq 0$, then $(A - uu^\top)^{-1} = A^{-1} + A^{-1}uu^\top A^{-1}/c$.
(b) Show that $h_{ii} < 1$ and $y_i - x_i^\top \hat\beta_{(i)} = (y_i - \hat y_i)/(1 - h_{ii})$.

Solution [Supplement].

(a) Multiplying out,

$$\begin{aligned}(A - uu^\top)\Big(A^{-1} + \frac{A^{-1}uu^\top A^{-1}}{c}\Big) &= I + \frac{uu^\top A^{-1}}{c} - uu^\top A^{-1} - \frac{u(u^\top A^{-1}u)u^\top A^{-1}}{c} \\ &= I + uu^\top A^{-1}\,\frac{1 - c - u^\top A^{-1}u}{c} = I.\end{aligned}$$

(b)

Step 1 ($h_{ii} < 1$). If $x_i = 0$ then $h_{ii} = 0$. Otherwise put $v = A^{-1}x_i \neq 0$. For every w, $w^\top A w = \|Xw\|^2 + \lambda\|w\|^2 \geq (x_i^\top w)^2 + \lambda\|w\|^2$. With $w = v$: $v^\top A v = x_i^\top A^{-1}x_i = h_{ii}$ and $x_i^\top v = h_{ii}$, so $h_{ii} \geq h_{ii}^2 + \lambda\|v\|^2 > h_{ii}^2$, i.e. $0 \leq h_{ii} < 1$.

Step 2. Deleting row i gives $A_{(i)} = A - x_i x_i^\top$ and $\hat\beta_{(i)} = A_{(i)}^{-1}(X^\top y - x_i y_i)$. By (a) with $u = x_i$ and $c = 1 - h_{ii} > 0$,

$$x_i^\top A_{(i)}^{-1} = x_i^\top A^{-1} + \frac{h_{ii}\, x_i^\top A^{-1}}{1 - h_{ii}} = \frac{x_i^\top A^{-1}}{1 - h_{ii}}.$$

Step 3. Therefore $x_i^\top \hat\beta_{(i)} = \dfrac{x_i^\top A^{-1}X^\top y - h_{ii}y_i}{1 - h_{ii}} = \dfrac{\hat y_i - h_{ii}y_i}{1 - h_{ii}}$, and

$$y_i - x_i^\top \hat\beta_{(i)} = \frac{(1 - h_{ii})y_i - \hat y_i + h_{ii}y_i}{1 - h_{ii}} = \frac{y_i - \hat y_i}{1 - h_{ii}}.$$

$\lambda = 0$ is the least-squares case, provided $A_{(i)}$ is invertible. (With re-centering/re-standardizing inside each fold the identity is only approximate.) ■

Exercise 3.3 (Bootstrap standard error of the mean)

Fix data $x_1, \ldots, x_n$ and let $x_1^*, \ldots, x_n^*$ be drawn i.i.d. from the empirical distribution $\hat P_n$, with mean $\bar x^*$.
(a) Show that $\mathbb{E}^*\bar x^* = \bar x$ and $\mathrm{Var}^*\bar x^* = \hat\sigma^2/n$, where $\hat\sigma^2 = \frac{1}{n}\sum_i (x_i - \bar x)^2$.
(b) Show that $\mathrm{SE}_B \to \hat\sigma/\sqrt{n} = \sqrt{(n-1)/n}\; s/\sqrt{n}$ almost surely as $B \to \infty$, with s^2 the usual sample variance.

Solution [Supplement].

(a) Each x_j^* takes the value x_i with probability $1/n$, so $\mathbb{E}^* x_j^* = \bar{x}$ and $\mathrm{Var}^* x_j^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$. The mean of n i.i.d. copies has mean $\bar{x}$ and variance $\hat{\sigma}^2/n$.

(b) Conditionally on the data, the bootstrap replicates $\bar{x}^{*1}, \dots, \bar{x}^{*B}$ are i.i.d. and bounded. By the strong law of large numbers, $\frac{1}{B}\sum_r \bar{x}^{*r} \to \bar{x}$ and $\frac{1}{B}\sum_r (\bar{x}^{*r})^2 \to \hat{\sigma}^2/n + \bar{x}^2$ almost surely, so

$$\mathrm{SE}_B^2 = \frac{B}{B-1}\Big[\frac{1}{B}\sum_r (\bar{x}^{*r})^2 - \Big(\frac{1}{B}\sum_r \bar{x}^{*r}\Big)^2\Big] \to \frac{\hat{\sigma}^2}{n}.$$

Since $\hat{\sigma}^2 = (n-1)s^2/n$, the limit is $\sqrt{(n-1)/n}\, s/\sqrt{n}$. ■

Chapter 4. Linear Regression

Lecture 6-1 (simple and multiple regression, inference, F-test, qualitative predictors, Gauss–Markov, interactions, polynomials)

1 Simple Linear Regression

True regression functions are never exactly linear, yet the linear model is very useful. Advertising data: is budget related to sales, how strongly, and which media contribute?

1.1 Model and least squares

$$Y = \beta_0 + \beta_1 X + \varepsilon, \qquad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

With residuals $e_i = y_i - \hat{y}_i$, $\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

Theorem: least-squares estimates

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Proof [Supplement]. $\partial \mathrm{RSS}/\partial \beta_0 = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ gives $\beta_0 = \bar{y} - \beta_1 \bar{x}$. Substituting, $\partial \mathrm{RSS}/\partial \beta_1 = -2\sum(x_i - \bar{x})\{(y_i - \bar{y}) - \beta_1(x_i - \bar{x})\} = 0$ (using $\sum(x_i - \bar{x}) = 0$) gives $\hat{\beta}_1$. RSS is a convex quadratic, so this is the minimum. ■

Advertising: $\widehat{\texttt{sales}} = 7.03 + 0.0475 \times \texttt{TV}$.

1.2 Accuracy of the coefficient estimates

Lines fitted to repeated samples from $f(X) = 2 + 3X$ scatter around it; the standard error measures this:

Theorem: standard errors

If the errors are uncorrelated with common variance $\sigma^2 = \mathrm{Var}(\varepsilon)$ and the x_i are fixed,

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2\Big[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\Big].$$

Proof [Supplement]. $\hat{\beta}_1 = \sum_i w_i y_i$ with $w_i = (x_i - \bar{x})/S_{xx}$, $S_{xx} = \sum(x_i - \bar{x})^2$; so $\mathrm{Var}\hat{\beta}_1 = \sigma^2 \sum w_i^2 = \sigma^2/S_{xx}$. Next $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \frac{\sigma^2}{n}\sum w_i = 0$, so $\mathrm{Var}\hat{\beta}_0 = \sigma^2/n + \bar{x}^2\sigma^2/S_{xx}$. ■

σ is estimated by the **residual standard error** $\mathrm{RSE} = \sqrt{\mathrm{RSS}/(n-2)}$; large S_{xx} gives a precise slope.

1.3 Confidence intervals and tests

A $100(1-\alpha)\%$ confidence interval covers the true parameter with probability $1-\alpha$ under repeated sampling:

$$\hat{\beta}_1 \pm t_{1-\alpha/2,\,n-2} \cdot \mathrm{SE}(\hat{\beta}_1) \quad (\approx \hat{\beta}_1 \pm 2\,\mathrm{SE}(\hat{\beta}_1) \text{ for } 95\%).$$

Advertising: $[0.042, 0.053]$.

$H_0 : \beta_1 = 0$ (no relationship) versus $H_A : \beta_1 \neq 0$:

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0 \text{ (Gaussian errors)},$$

p-value $= P(|T| \geq |t|)$.

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

1.4 Overall accuracy: RSE and R^2

$\mathrm{RSE} = \sqrt{\frac{1}{n-2}\sum(y_i - \hat{y}_i)^2}$ estimates $\mathrm{sd}(\varepsilon)$ (lack of fit, in units of Y);

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \mathrm{TSS} = \sum(y_i - \bar{y})^2,$$

the fraction of variance explained.

Theorem: $R^2 = r^2$ in simple regression

With $r = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$, simple linear regression has $R^2 = r^2$.

Proof [Supplement]. $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, and $\mathrm{TSS} = \mathrm{RSS} + \sum(\hat{y}_i - \bar{y})^2$ (normal equations). So $R^2 = \hat{\beta}_1^2 S_{xx}/S_{yy} = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

Sales on TV: RSE $= 3.26$ (about 3,260 units), $R^2 = 0.612$, $F = 312.1$.

2 Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon, \qquad \texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \beta_2 \texttt{radio} + \beta_3 \texttt{newspaper} + \varepsilon.$$

β_j: average effect of a one-unit increase in X_j, *holding all other predictors fixed.*

2.1 Interpreting coefficients

Correlated predictors make interpretation hazardous and inflate all coefficient variances; uncorrelated (balanced) designs avoid this. Avoid causal claims for observational data.

Example: the woes of interpreting coefficients (Mosteller and Tukey, 1977)

Y = money in your pocket, X_1 = number of coins, X_2 = number of pennies, nickels and dimes. Alone, X_2 is positively associated with Y; given X_1, more small coins means fewer quarters, so its coefficient is negative.

2.2 Estimation

$\hat{\beta}$ minimizes $\mathrm{RSS} = \sum_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2$: $\hat{\beta} = (X^\top X)^{-1} X^\top y$ (normal equations $X^\top(y - X\hat{\beta}) = 0$). For $p = 2$ the fit is a plane.

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

An extra \$1,000 on radio: about 189 more units at fixed TV and newspaper. Newspaper is significant alone but not here: it is correlated with radio and acts as a surrogate.

3 Four Important Questions

3.1 Is at least one predictor useful?

$H_0 : \beta_1 = \cdots = \beta_p = 0$ versus H_A: some $\beta_j \neq 0$,

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)} \;\sim\; F_{p,\,n-p-1} \text{ under } H_0.$$

Derivation: why F is the right statistic

$\mathbb{E}[\text{RSS}/(n-p-1)] = \sigma^2$, and under H_0 also $\mathbb{E}[(\text{TSS}-\text{RSS})/p] = \sigma^2$ (Exercise 4.1); otherwise the numerator is larger. With Gaussian errors, $\text{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$ and $(\text{TSS}-\text{RSS})/\sigma^2 \sim \chi^2_p$ independently under H_0 (Cochran). With $p = 100$ and H_0 true, about 5 individual t-tests have $p < 0.05$ by chance; the F-test keeps its level.

Advertising: RSE $= 1.69$, $R^2 = 0.897$, $F = 570$.

3.2 Deciding on the important variables

Best subsets: 2^p models (over a billion for $p = 40$). **Forward selection** adds, one at a time, the most improving variable; **backward elimination** removes the least useful one. Each searches $1 + p(p+1)/2$ models (Ch. 6).

3.3 How well does the model fit?

$\text{RSE} = \sqrt{\text{RSS}/(n-p-1)}$, $R^2 = 1 - \text{RSS}/\text{TSS}$, residual plots. R^2 never decreases with added predictors; RSE can increase.

3.4 How accurate is the prediction?

(i) **reducible error**: $\hat{Y} = \hat{\beta}_0 + \cdots + \hat{\beta}_p X_p$ estimates $f(X) = \beta_0 + \cdots + \beta_p X_p$ (confidence interval);

(ii) **model bias**: the linear model is an approximation;

(iii) **irreducible error** ε (prediction interval, always wider). For a new x_0: $\text{Var}(x_0^\top \hat{\beta}) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0$ and $\text{Var}(Y_0 - x_0^\top \hat{\beta}) = \sigma^2\{1 + x_0^\top (X^\top X)^{-1} x_0\}$.

4 Qualitative Predictors

Categorical predictors (factors), e.g. student or marital status in the Credit data.

4.1 Two levels

$x_i = 1$ if person i owns a house, 0 otherwise:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{owner,} \\ \beta_0 + \varepsilon_i & \text{non-owner.} \end{cases}$$

Intercept 509.80 (SE 33.13), own[Yes] 19.73 (SE 46.05, $p = 0.669$): average debt \$509.80 versus \$529.53, not significantly different.

4.2 More than two levels

Region (East, South, West): $x_{i1} = \mathbb{I}(\text{South})$, $x_{i2} = \mathbb{I}(\text{West})$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{South,} \\ \beta_0 + \beta_2 + \varepsilon_i & \text{West,} \\ \beta_0 + \varepsilon_i & \text{East (baseline).} \end{cases}$$

One fewer dummy than levels (K dummies are collinear with the intercept). East \$531.00; South -18.69 ($p = 0.774$); West -12.50 ($p = 0.826$). Fitted values and the F-test do not depend on the baseline (Exercise 4.2).

5 The Gauss–Markov Theorem

Motivation. Under uncorrelated equal-variance errors, least squares has the smallest variance among linear unbiased estimators, without normality.

Theorem: Gauss–Markov

Let $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and X of full column rank. For any fixed c, among all linear unbiased estimators of $\theta = c^\top \beta$, the estimator $\hat{\theta} = c^\top \hat{\beta}$, $\hat{\beta} = (X^\top X)^{-1} X^\top y$, has the minimum variance, and it is unique. Least squares is the **BLUE**.

Proof [Supplement]. *Step 1 (unbiasedness constraint).* For $\tilde{\theta} = a^\top y$, $\mathbb{E}(a^\top y) = a^\top X\beta = c^\top \beta$ for all β iff $X^\top a = c$.

Step 2 (variance comparison). Least squares is $a_0 = X(X^\top X)^{-1} c$, with $X^\top a_0 = c$. Any unbiased $a = a_0 + d$ has $X^\top d = 0$, so $a_0^\top d = c^\top (X^\top X)^{-1} X^\top d = 0$ and

$$\mathrm{Var}(\tilde{\theta}) = \sigma^2 a^\top a = \sigma^2 (a_0^\top a_0 + d^\top d) \;\geq\; \sigma^2 a_0^\top a_0 = \mathrm{Var}(\hat{\theta}),$$

with equality iff $d = 0$. ■

Biased estimators such as ridge (Ch. 6) can have smaller mean squared error.

6 Extensions of the Linear Model

The standard model assumes **additive** and **linear** effects.

6.1 Interactions

If radio makes TV advertising more effective (*synergy*, or *interaction*), the TV effect is not a constant β_1. Advertising residuals: sales are overestimated when either budget is low and underestimated when the budget is split.

$$\texttt{sales} = \beta_0 + \beta_1\texttt{TV} + \beta_2\texttt{radio} + \beta_3(\texttt{radio} \times \texttt{TV}) + \varepsilon.$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

R^2: 89.7% → 96.8%, i.e. $(96.8 - 89.7)/(100 - 89.7) = 69\%$ of the remaining variability. An extra \$1,000 of TV: $(\hat{\beta}_1 + \hat{\beta}_3\,\texttt{radio}) \times 1000 = 19 + 1.1\,\texttt{radio}$ units; of radio: $29 + 1.1\,\texttt{TV}$ units.

Hierarchy principle. With an interaction, include the main effects even if not significant; omitting them changes the meaning of the interaction (and makes it depend on the origin of the predictors).

6.2 Qualitative–quantitative interactions

$$\texttt{balance}_i \approx \beta_0 + \beta_1\texttt{income}_i + \begin{cases} \beta_2 + \beta_3\texttt{income}_i & \text{student} \\ 0 & \text{non-student} \end{cases}$$

$$= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\texttt{income}_i & \text{student,} \\ \beta_0 + \beta_1\texttt{income}_i & \text{non-student :} \end{cases}$$

different intercepts *and* slopes.

6.3 Nonlinear effects: polynomial regression

Auto data: $\texttt{mpg} = \beta_0 + \beta_1\texttt{hp} + \beta_2\texttt{hp}^2 + \varepsilon$, estimates 56.9001, −0.4662, 0.0012 (all $p < 0.0001$). Still *linear* in the coefficients. Residuals versus fitted: strong U-shape for the linear fit, little pattern for the quadratic.

Summary

- OLS minimizes RSS; SEs, t-tests and CIs quantify coefficient uncertainty; RSE and R^2 measure fit.
- F-test: is any predictor useful; dummies encode factors relative to a baseline.
- Gauss–Markov: OLS is BLUE. Interactions and polynomials stay within least squares.

7 Selected exercises

Motivation. The expectations behind F, baseline invariance of dummy coding, and ISLR Ex. 3.3.

Exercise 4.1 (Expected residual and regression sums of squares)

Let $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2 I_n$, and $X \in \mathbb{R}^{n\times(p+1)}$ of full column rank with first column $\mathbf{1}$.
(a) Show that $\mathbb{E}[\mathrm{RSS}] = (n-p-1)\sigma^2$, so RSE^2 is unbiased for σ^2.
(b) Show that under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $\mathbb{E}[\mathrm{TSS} - \mathrm{RSS}] = p\sigma^2$.

Solution [Supplement].

(a)

Step 1. $H = X(X^\top X)^{-1}X^\top$ is a symmetric idempotent with $HX = X$, so $(I-H)y = (I-H)\varepsilon$ and $\mathrm{RSS} = \varepsilon^\top(I-H)\varepsilon$.

Step 2. For fixed M, $\mathbb{E}[\varepsilon^\top M\varepsilon] = \mathbb{E}\mathrm{tr}(M\varepsilon\varepsilon^\top) = \sigma^2\mathrm{tr}M$.

Step 3. $\mathrm{tr}(I-H) = n - \mathrm{tr}((X^\top X)^{-1}X^\top X) = n-(p+1)$.

(b) $J = \mathbf{1}\mathbf{1}^\top/n$ projects onto $\mathrm{span}(\mathbf{1}) \subseteq \mathrm{col}(X)$, so $HJ = JH = J$ and $H-J$ is a projection of trace p. $\mathrm{TSS}-\mathrm{RSS} = y^\top(I-J)y - y^\top(I-H)y = y^\top(H-J)y$. Under H_0, $y = \beta_0\mathbf{1}+\varepsilon$ and $(H-J)\mathbf{1} = 0$, so this is $\varepsilon^\top(H-J)\varepsilon$, with expectation $p\sigma^2$ by Step 2. ■

Exercise 4.2 (The baseline does not matter)

A factor has levels $1,\dots,K$, each observed at least once, with indicator columns $D_1,\dots,D_K \in \mathbb{R}^n$. Design A uses $\mathbf{1}, D_2,\dots,D_K$ (baseline 1); design B uses $\mathbf{1}$ and all D_k with $k \neq r$ (baseline r); other predictors, if any, are the same in both.
(a) Show that A and B have the same column space, hence the same fitted values, RSS and overall F-statistic.
(b) Express the coefficients of B through those of A.

Solution [Supplement].

(a) $\sum_{k=1}^K D_k = \mathbf{1}$, so $D_1 = \mathbf{1} - \sum_{k\geq 2} D_k \in \mathrm{col}(A)$ and $\mathrm{col}(A) = \mathrm{span}(D_1,\dots,D_K,\text{others})$; likewise for $\mathrm{col}(B)$. Equal column spaces give the same projection, $\hat{y}$ and RSS; TSS, n and the number of predictors agree, so F agrees.

(b) With no other predictors, set $\beta_1^A := 0$, $\beta_r^B := 0$. By (a), $\beta_0^A + \beta_k^A = \beta_0^B + \beta_k^B$ for each level k (coefficients are unique by full column rank). $k = r$ gives $\beta_0^B = \beta_0^A + \beta_r^A$, hence $\beta_k^B = \beta_k^A - \beta_r^A$. Credit example with baseline South: intercept 512.31, East +18.69, West +6.19. ■

Exercise 4.3 (Reading an interaction model; ISLR Ex. 3.3)

Starting salary (thousands of dollars) is regressed on $X_1 = \mathrm{GPA}$, $X_2 = \mathrm{IQ}$, $X_3 = \mathrm{Level}$ (1 college, 0 high school), $X_4 = X_1X_2$ and $X_5 = X_1X_3$, giving $\hat{\beta}_0 = 50$, $\hat{\beta}_1 = 20$, $\hat{\beta}_2 = 0.07$, $\hat{\beta}_3 = 35$, $\hat{\beta}_4 = 0.01$, $\hat{\beta}_5 = -10$.
(a) For fixed IQ and GPA, who earns more on average, college or high-school graduates?
(b) Predict the salary of a college graduate with IQ 110 and GPA 4.0.
(c) True or false: since $\hat{\beta}_4$ is very small, there is little evidence of a GPA–IQ interaction.

Solution.

(a) College minus high school: $\hat{\beta}_3 + \hat{\beta}_5\,\mathrm{GPA} = 35 - 10\,\mathrm{GPA}$, positive iff $\mathrm{GPA} < 3.5$. High-school graduates earn more if $\mathrm{GPA} > 3.5$; below that, college graduates.

(b) $50 + 20(4) + 0.07(110) + 35 + 0.01(4)(110) - 10(4) = 50 + 80 + 7.7 + 35 + 4.4 - 40 = 137.1$, about \$137,100.

(c) False. $\mathrm{GPA}\times\mathrm{IQ} \approx 400$, so 0.01 per unit is worth about 4 thousand dollars; evidence is measured

by $\hat{\beta}_4/\text{SE}(\hat{\beta}_4)$, which is not given. ■

Chapter 5. Potential Problems in Regression

Lecture 6-2 (assumptions, non-Gaussianity, outliers and robust regression, breakdown point, measurement error)

1 Assumptions and Common Issues

Assumptions: linearity; normal, independent, constant-variance errors; no outliers (large residuals) or high-leverage points (unusual predictors); small collinearity.

Kind of problem	Examples
data problems	outliers, measurement error, missing data
model problems	omitted-variable bias, misspecification, multicollinearity
error problems	non-normality, heteroscedasticity, autocorrelation

2 Non-Gaussianity: Normality Is Not an Issue

Motivation. Estimates are unbiased without normality and the CLT makes inference approximately valid; independence and constant variance matter more.

Theorem: exact distribution under Gaussian errors

If $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$ and X of full column rank, then

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1}), \qquad \hat{\beta}_j \sim N(\beta_j, \sigma^2 v_j),$$

where v_j is the jth diagonal element of $(X^\top X)^{-1}$. Replacing σ by $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p)}$ (p columns of X including the intercept),

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p},$$

which gives tests and confidence intervals.

Proof [Supplement]. $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ is a linear map of a Gaussian vector, with mean β and covariance $\sigma^2 (X^\top X)^{-1}$. The residual vector $(I - H)\varepsilon$ is independent of $\hat{\beta}$ (uncorrelated jointly Gaussian), and $\mathrm{RSS}/\sigma^2 = \varepsilon^\top (I - H)\varepsilon / \sigma^2 \sim \chi^2_{n-p}$ since $I - H$ is a projection of rank $n - p$. The ratio of a standard normal and an independent $\sqrt{\chi^2_{n-p}/(n-p)}$ is t_{n-p}. ■

2.1 Large-sample inference

By the CLT,

$$\hat{\beta} \;\dot{\sim}\; N(\beta, \sigma^2 (X^\top X)^{-1}) \quad \text{for large } n,$$

for i.i.d. errors with finite variance and $\max_i h_{ii} \to 0$; t-tests and intervals remain approximately valid. Alternatively, **bootstrap inference** (Ch. 3); resampling pairs (x_i, y_i) is also valid under heteroscedasticity.

3 Outliers and Robust Regression

3.1 Why OLS is sensitive

Squaring residuals in $\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2$ amplifies large errors, so one extreme observation can dominate the fit.

3.2 Alternative objectives

- **Least absolute deviation (LAD)**: $\hat{\beta} = \arg\min_\beta \sum_i |y_i - x_i^\top \beta|$; targets the conditional median (Exercise 2.1).
- **Huber regression**: $\hat{\beta} = \arg\min_\beta \sum_i \rho_c(r_i)$, $r_i = y_i - x_i^\top \beta$,

$$\rho_c(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le c, \\ c|r| - \frac{1}{2}c^2, & |r| > c. \end{cases}$$

- **Least trimmed squares (LTS)**: $\hat{\beta} = \arg\min_\beta \sum_{i=1}^{h} r^2_{(i)}$, using only the h smallest squared residuals ($r^2_{(1)} \le r^2_{(2)} \le \cdots$).

Derivation: Huber as iteratively reweighted least squares

ρ_c is convex with derivative $\psi_c(r) = r$ for $|r| \le c$ and $c\,\mathrm{sign}(r)$ otherwise. The estimating equation $\sum_i \psi_c(r_i)x_i = 0$ equals the weighted normal equation $\sum_i w_i r_i x_i = 0$ with $w_i = \min(1, c/|r_i|)$, so iterate $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$. Influence is bounded by c (unbounded under OLS). Usually $c = 1.345\hat{s}$, $\hat{s}$ a robust scale (95% efficiency under normality).

3.3 Breakdown point

Definition: breakdown point

The (finite-sample) breakdown point is the smallest fraction of contaminated observations that can make the estimate arbitrarily large.

OLS	LAD	Huber M-estimator	least median of squares	least trimmed squares
0	0	0	0.5	≈ 0.5

OLS: breakdown $1/n \to 0$ (Exercise 5.1). LAD and Huber resist outliers in y but not a bad *leverage point*. LMS (minimize the median squared residual) and LTS with $h \approx n/2$ tolerate nearly 50%, the maximum possible.

4 Measurement Error

True model $Y = \beta_0 + \beta_1 X + \varepsilon$, but

$$\tilde{X} = X + u, \qquad u \text{ independent of } (X, \varepsilon),\ \mathbb{E}u = 0.$$

Regressing Y on $\tilde{X}$:

Theorem: attenuation bias

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \frac{\mathrm{Var}(X)}{\mathrm{Var}(X)+\mathrm{Var}(u)} = \lambda\beta_1, \qquad 0<\lambda\leq 1.$$

Proof [Supplement]. The OLS slope converges to $\mathrm{Cov}(\tilde{X},Y)/\mathrm{Var}(\tilde{X})$. Here $\mathrm{Cov}(X+u, \beta_0+\beta_1 X+\varepsilon) = \beta_1 \mathrm{Var}X$ and $\mathrm{Var}(\tilde{X}) = \mathrm{Var}X + \mathrm{Var}u$. ■

Attenuation bias toward zero; λ is the reliability ratio. Error in Y only inflates the error variance (Exercise 5.2). In multiple regression, error in one predictor also biases correlated predictors' coefficients.

4.1 Remedies

Method	Idea
instrumental variables	Z correlated with X, independent of the error: $\hat{\beta}_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z,Y)/\widehat{\mathrm{Cov}}(Z,\tilde{X})$
repeated measurements	averaging m replicates gives $\mathrm{Var}u/m$; replicates estimate $\mathrm{Var}u$, and $\hat{\beta}_1/\hat{\lambda}$ corrects the bias
errors-in-variables models	model the measurement process (e.g. Deming regression)
SIMEX	add noise of increasing size and extrapolate back to zero error
validation data	calibrate $\tilde{X}$ on a subset with true X (regression calibration)

Summary

- Normality is often not critical (CLT, bootstrap).
- Robust regression (LAD, Huber, LTS); only high-breakdown methods resist leverage points.
- Predictor error attenuates the slope; IV, replicates, SIMEX or validation data correct it.

5 Selected exercises

Motivation. One outlier breaks OLS; the IV estimator is consistent.

Exercise 5.1 (One outlier breaks OLS)

Let $X \in \mathbb{R}^{n\times q}$ have full column rank q and first column $\mathbf{1}$, with rows $x_i^\top$, and let $H = X(X^\top X)^{-1}X^\top$ with diagonal entries h_{ii}. Replace y_i by $y_i + t$. Show that the new estimate is $\hat{\beta}(t) = \hat{\beta} + t(X^\top X)^{-1}x_i$, that $\|\hat{\beta}(t)\| \to \infty$ as $|t| \to \infty$, and that the fitted value at x_i moves by $t\,h_{ii}$ with $h_{ii} > 0$.

Solution [Supplement]. $\hat{\beta}(t) = (X^\top X)^{-1}X^\top(y + te_i) = \hat{\beta} + t(X^\top X)^{-1}x_i$. $u = (X^\top X)^{-1}x_i \neq 0$ since $(X^\top X)^{-1}$ is invertible and $x_i \neq 0$ (first entry 1), so $\|\hat{\beta}(t)\| \geq |t|\,\|u\| - \|\hat{\beta}\| \to \infty$. The fitted value at x_i changes by $x_i^\top t u = t\,h_{ii}$, with $h_{ii} = x_i^\top(X^\top X)^{-1}x_i > 0$ by positive definiteness. Hence the breakdown point of OLS is $1/n$ (cf. Exercise 1.3). ■

Exercise 5.2 (Error in Y, and instrumental variables)

Let $(X_i, Y_i, \dots)$ be i.i.d. with finite second moments, $Y = \beta_0 + \beta_1 X + \varepsilon$, $\mathrm{Var}X > 0$, $\mathbb{E}\varepsilon = 0$, $\mathrm{Cov}(X, \varepsilon) = 0$.
(a) If only $\tilde{Y} = Y + v$ is observed, with $\mathbb{E}v = 0$ and $\mathrm{Cov}(X, v) = 0$, show that the OLS slope of $\tilde{Y}$ on X is consistent for β_1.
(b) If only $\tilde{X} = X + u$ is observed (as in the attenuation theorem) and Z satisfies $\mathrm{Cov}(Z, X) \neq 0$, $\mathrm{Cov}(Z, u) = \mathrm{Cov}(Z, \varepsilon) = 0$, show that $\hat{\beta}_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z, Y)/\widehat{\mathrm{Cov}}(Z, \tilde{X}) \xrightarrow{p} \beta_1$.

Solution [Supplement]. By the LLN every sample covariance converges in probability to its population value; ratios converge when the denominator limit is nonzero (continuous mapping).

(a) $\widehat{\mathrm{Cov}}(X, \tilde{Y})/\widehat{\mathrm{Var}}(X) \xrightarrow{p} \mathrm{Cov}(X, \beta_0 + \beta_1 X + \varepsilon + v)/\mathrm{Var}X = \beta_1$; the error $\varepsilon + v$ only raises the noise level.

(b) $\mathrm{Cov}(Z, Y) = \beta_1 \mathrm{Cov}(Z, X) + \mathrm{Cov}(Z, \varepsilon) = \beta_1 \mathrm{Cov}(Z, X)$ and $\mathrm{Cov}(Z, \tilde{X}) = \mathrm{Cov}(Z, X) + \mathrm{Cov}(Z, u) = \mathrm{Cov}(Z, X) \neq 0$; the ratio is β_1. ■

Chapter 6. Explainable Models: Subset Selection, Shrinkage and Dimension Reduction

Lecture 6-3 (best subset, stepwise, C_p/AIC/BIC/CV, ridge, lasso, high-dimensional theory, PCR, PLS)

1 Three Classes of Methods

- **Subset selection**: least squares on a subset of the p predictors.
- **Shrinkage (regularization)**: shrink all coefficients toward zero; reduces variance, may select variables.
- **Dimension reduction**: least squares on $M < p$ linear projections of the predictors.

2 Subset Selection

Motivation. How to search the 2^p subsets and compare models of different size.

2.1 Best subset selection

Algorithm: best subset selection

1. Let $\mathcal{M}_0$ be the null model (no predictors; predicts the sample mean).
2. For $k = 1, \ldots, p$: fit all $\binom{p}{k}$ models with exactly k predictors and call the best (smallest RSS, or largest R^2) $\mathcal{M}_k$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using a chosen score (CV error, C_p, AIC, BIC, adjusted R^2).

Across sizes RSS always favors the largest model, so step 3 needs a test-error score. 2^p models: infeasible for large p, and the huge search space invites overfitting.

2.2 Stepwise selection

Algorithm: forward stepwise selection

1. Let $\mathcal{M}_0$ be the null model.
2. For $k = 0, \ldots, p-1$: consider the $p-k$ models that add one predictor to $\mathcal{M}_k$; the best of them is $\mathcal{M}_{k+1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ by CV error, C_p, AIC, BIC or adjusted R^2.

Algorithm: backward elimination

1. Let $\mathcal{M}_p$ be the full model.
2. For $k = p, p-1, \ldots, 1$: consider the k models that drop one predictor from $\mathcal{M}_k$; the best is $\mathcal{M}_{k-1}$.

3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$.

Both search $1 + \sum_{k=0}^{p-1}(p-k) = 1 + p(p+1)/2$ models (211 versus $2^{20} \approx 10^6$ for $p = 20$), without guarantee: the best two-variable model need not contain the best single predictor. Backward elimination needs $n > p$ ($X^\top X$ invertible); forward selection works for $n < p$ (up to $\mathcal{M}_{n-1}$).

2.3 Choosing the optimal model

Indirect test-error estimates adjust the training error:

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \text{AIC, BIC (Ch. 2)}, \qquad R^2_{\mathrm{adj}},$$

with d the number of parameters and $\hat{\sigma}^2$ estimated from the full model.

Theorem: C_p is unbiased for the in-sample prediction error

For a least-squares fit $\hat{y} = Hy$ with $\mathrm{tr}H = d$, $y = \mu + \varepsilon$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and new responses y^{new} at the same inputs,

$$\mathbb{E}\Big[\frac{1}{n}\|y^{\mathrm{new}} - \hat{y}\|^2\Big] = \mathbb{E}\Big[\frac{1}{n}\|y - \hat{y}\|^2\Big] + \frac{2d\sigma^2}{n}.$$

Proof [Supplement]. Expanding both sides, the difference is $\frac{2}{n}\sum_i \mathrm{Cov}(y_i, \hat{y}_i) = \frac{2}{n}\mathrm{trCov}(y, Hy) = \frac{2}{n}\sigma^2 \mathrm{tr}H = \frac{2d\sigma^2}{n}$. ■

Directly, validation or CV error needs fewer assumptions and no $\hat{\sigma}^2$. Credit data: $\sqrt{\mathrm{BIC}}$, validation and CV error flatten after four to six predictors. **One-standard-error rule**: the smallest model within one SE of the minimum.

3 Shrinkage Methods

Fit all p predictors with coefficients shrunk toward zero, reducing their variance.

3.1 Ridge regression

$$\hat{\beta}^R = \arg\min_{\beta} \Big\{ \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij}\Big)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \Big\} = \arg\min_{\beta} \{\mathrm{RSS} + \lambda \|\beta\|_2^2\},$$

$\lambda \geq 0$ chosen by CV, intercept not penalized; $\lambda = 0$ is least squares, $\lambda \to \infty$ gives $\hat{\beta}^R \to 0$. For centered data $\hat{\beta}^R = (X^\top X + \lambda I)^{-1} X^\top Y$.

Scaling. Least squares is scale equivariant ($X_j \to cX_j$ gives $\hat{\beta}_j \to \hat{\beta}_j/c$); ridge is not. Standardize first:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}}.$$

Credit data: standardized coefficients shrink smoothly to zero as λ grows.

3.2 Why ridge improves over least squares

Simulation, $n = 50$, $p = 45$, all coefficients nonzero: test MSE is minimized at an intermediate λ, well below least squares.

Theorem: bias and variance of ridge

For $Y = X\beta + \varepsilon$, $\varepsilon \sim (0, \sigma^2 I)$, and $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top Y$,

$$\mathbb{E}\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top X \beta, \qquad \text{Bias} = -\lambda (X^\top X + \lambda I)^{-1}\beta,$$

$$\mathrm{Var}(\hat\beta_\lambda) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}, \qquad \mathrm{MSE} = \|\text{Bias}\|^2 + \mathrm{trVar}.$$

Moreover there is always some $\lambda > 0$ with $\mathrm{MSE}(\hat\beta_\lambda) < \mathrm{MSE}(\hat\beta_{\mathrm{OLS}})$ (Hoerl and Kennard, 1970).

Proof [Supplement]. The mean and variance follow from linearity in Y; $(X^\top X + \lambda I)^{-1} X^\top X - I = -\lambda (X^\top X + \lambda I)^{-1}$. With $X^\top X = V D^2 V^\top$ and $\theta = V^\top \beta$,

$$\mathrm{MSE}(\lambda) = \sum_{j=1}^{p} \frac{\sigma^2 d_j^2 + \lambda^2 \theta_j^2}{(d_j^2 + \lambda)^2}.$$

Differentiating, $\mathrm{MSE}'(\lambda) = \sum_j \frac{2 d_j^2 (\lambda \theta_j^2 - \sigma^2)}{(d_j^2 + \lambda)^3}$, which equals $-2\sigma^2 \sum_j d_j^{-4} < 0$ at $\lambda = 0$, so MSE decreases for small $\lambda > 0$. ∎

The gain is largest when some d_j are small (collinearity) or $p \approx n$.

3.3 The lasso

Ridge keeps all p predictors; the lasso

$$\hat\beta^L = \arg\min_\beta \{\mathrm{RSS} + \lambda \|\beta\|_1\}, \qquad \|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

sets some coefficients **exactly zero** for large λ: variable selection, **sparse** models. Credit data: coefficients enter the path one at a time as λ decreases.

3.4 Constrained formulations and geometry

$$\min_\beta \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_j \beta_j x_{ij}\Big)^2 \quad \text{subject to} \quad \sum_j \beta_j^2 \le s \quad \text{resp.} \quad \sum_j |\beta_j| \le s,$$

for some $s = s(\lambda)$ (Lagrange duality); best subset uses $\sum_j \mathbb{I}(\beta_j \neq 0) \le s$.

Theorem: orthonormal design

If $n = p$ and $X = I$ (no intercept), then for RSS + λ·penalty:

$$\hat\beta_j^R = \frac{y_j}{1 + \lambda}, \qquad \hat\beta_j^L = \begin{cases} y_j - \lambda/2, & y_j > \lambda/2, \\ y_j + \lambda/2, & y_j < -\lambda/2, \\ 0, & |y_j| \le \lambda/2. \end{cases}$$

Proof [Supplement]. The objective separates: ridge minimizes $(y_j - \beta_j)^2 + \lambda \beta_j^2$; lasso minimizes $(y_j - \beta_j)^2 + \lambda |\beta_j|$, whose subgradient condition $-2(y_j - \beta_j) + \lambda s_j = 0$, $s_j \in \partial |\beta_j|$, gives the soft-thresholding rule. ∎

Ridge shrinks proportionally; the lasso soft-thresholds.

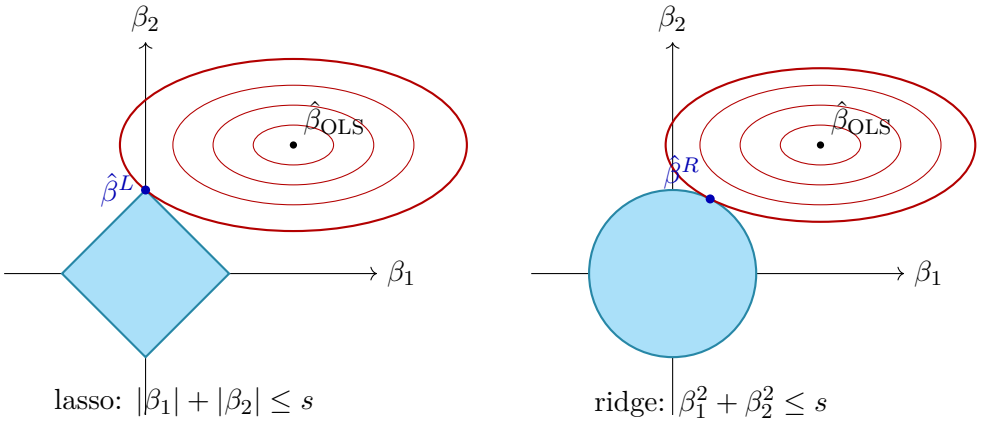


Figure 5: RSS contours and constraint regions. The diamond's corners lie on the axes, so the lasso often hits one ($\hat\beta_1 = 0$ here); the disc has none.

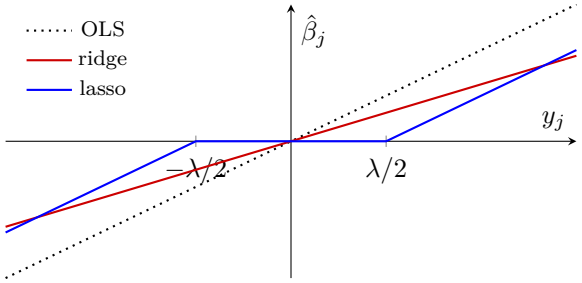


Figure 6: Orthonormal design ($\lambda = 0.6$).

3.5 Dense versus sparse truth

All 45 predictors matter: ridge slightly beats the lasso; only 2 of 45: the lasso wins clearly. The sparsity is unknown, so CV decides.

3.6 Selecting the tuning parameter

Grid of λ, CV error for each, minimizer, refit on all data.

4 Theoretical View in High Dimensions

$Y = X\beta + \varepsilon$, $X \in \mathbb{R}^{n\times p}$, possibly $p \gg n$, where OLS is unstable or not identifiable. **Ridge** assumes β dense but small in ℓ_2; the **lasso** assumes β (approximately) sparse.

Theorem: lasso rate under sparsity (informal)

Let β be s-sparse, $\|\beta\|_0 \le s$. Under a restricted-eigenvalue (RE) condition on X and with $\lambda \asymp \sigma\sqrt{\log p/n}$ (in the normalization $\frac{1}{2n}\text{RSS} + \lambda\|\beta\|_1$),

$$\|\hat\beta - \beta\|_2^2 \lesssim \frac{\sigma^2 s \log p}{n}$$

with high probability. Hence consistent estimation is possible when $n \gg s \log p$.

Not knowing the support costs only a $\log p$ factor over the oracle rate $\sigma^2 s/n$.

Why design assumptions are needed. With highly correlated columns, different sparse vectors

give similar fits. Guarantees need restricted eigenvalue, compatibility or restricted isometry; exact support recovery needs an irrepresentable condition and a beta-min condition ($\min_{j\in S}|\beta_j|$ above the noise level).

Ridge does not exploit sparsity but stabilizes estimation under multicollinearity; it suits dense β with small ℓ_2 norm.

Effective complexity. $B_0(s) = \{\beta : \|\beta\|_0 \le s\}$ has complexity about $s \log p$; $B_2(R) = \{\beta : \|\beta\|_2 \le R\}$ is governed by ℓ_2 geometry.

5 Dimension Reduction Methods

5.1 General idea

Construct $M < p$ linear combinations

$$Z_m = \sum_{j=1}^{p} \phi_{mj} X_j, \qquad m = 1, \dots, M,$$

and fit $Y = \theta_0 + \sum_{m=1}^{M} \theta_m Z_m + \varepsilon$ by least squares; the ϕ_{mj} define the method. Since

$$\sum_{m=1}^{M} \theta_m Z_m = \sum_{j=1}^{p} \beta_j X_j, \qquad \beta_j = \sum_{m=1}^{M} \theta_m \phi_{mj},$$

β is constrained to an M-dimensional subspace: less variance, bias if β lies outside it.

5.2 Principal components regression

Directions: principal components, i.e. successive maximal-variance uncorrelated combinations (eigenvectors of the sample covariance).

Example: advertising data, population and ad spending in 100 cities

$Z_1 = 0.839(\texttt{pop} - \overline{\texttt{pop}}) + 0.544(\texttt{ad} - \overline{\texttt{ad}})$, $0.839^2 + 0.544^2 = 1$: the line minimizing squared perpendicular distances. z_{i1} is strongly related to both pop and ad; second-PC scores only weakly.

Assumption: the directions of largest variation in X are those associated with Y. Sparse dependence on many predictors needs many components. $M = p$ is least squares; ridge shrinks low-variance directions smoothly, PCR drops them (Exercise 6.3). Credit data: CV error lowest near $M = 10$, little reduction.

Standardization: generally standardize before PCA, unless all variables share a unit.

Limitation: the directions are *unsupervised*, so they need not be best for predicting Y.

5.3 Partial least squares

PLS chooses $Z_1, \dots, Z_M$ *supervised*: approximating the predictors *and* related to the response.

Theorem: the first PLS direction

With standardized X and centered Y,

$$w_1 = \arg\max_{\|w\|=1} \mathrm{Cov}(Xw, Y)^2 \;\propto\; X^\top Y,$$

i.e. ϕ_{1j} is proportional to the simple-regression coefficient of Y on X_j; $Z_1 = Xw_1$.

Proof [Supplement]. $\mathrm{Cov}(Xw, Y) = \frac{1}{n} w^\top X^\top Y$, and by Cauchy–Schwarz $(w^\top X^\top Y)^2 \le \|w\|^2 \|X^\top Y\|^2$ with equality iff $w \propto X^\top Y$. For standardized X_j the simple-regression slope of Y on X_j is $x_{(j)}^\top Y/n$. ■

Algorithm: PLS (sequential construction)

Standardize X and Y; set $X^{(0)} = X$, $Y^{(0)} = Y$. For $m = 1, \dots, M$:

1. direction $w_m \propto X^{(m-1)\top} Y^{(m-1)}$;
2. component $Z_m = X^{(m-1)} w_m$;
3. regress and update residuals $Y^{(m)} = Y^{(m-1)} - \hat{\theta}_m Z_m$;
4. deflate the predictors $X^{(m)} = X^{(m-1)} - Z_m p_m^\top$, $p_m = X^{(m-1)\top} Z_m / (Z_m^\top Z_m)$ (each column is replaced by its residual after regressing on Z_m).

So $Y = \theta_1 Z_1 + E_1$, $E_1 = \theta_2 Z_2 + E_2$, …; the components are uncorrelated. Advertising: the first PLS direction weights `ad` more than the first PC. In practice PLS is often no better than ridge or PCR: less bias, possibly more variance.

5.4 Theoretical comparison

Method	Assumption on β	Mechanism
lasso	sparsity	ℓ_1 constraint
ridge	small ℓ_2 norm	shrinkage
PLS	low-dimensional structure	iterative projection

With $\Sigma = \mathrm{Cov}(X)$, $c = \mathrm{Cov}(X, Y)$, the m-component PLS estimator lies in the Krylov subspace

$$\mathcal{K}_m(\Sigma, c) = \mathrm{span}\{c, \Sigma c, \dots, \Sigma^{m-1} c\},$$

(least squares restricted to it, i.e. m conjugate-gradient steps on the normal equations); it is consistent if $\beta \in \mathcal{K}_m(\Sigma, c)$. Since $\beta = \Sigma^{-1} c$, this holds e.g. when c lies in the span of m eigenvectors of Σ.

Summary

- Stepwise search: $1 + p(p+1)/2$ models; size by C_p, AIC, BIC, adjusted R^2 or CV.
- Ridge shrinks (beats OLS for some λ); the lasso shrinks and selects ($s \log p/n$ under design conditions).
- PCR: unsupervised components; PLS: supervised directions (Krylov subspace).

6 Selected exercises

Motivation. ISLR Ex. 6.5 and 6.7 (duplicated predictors, Bayesian view) and ridge in the singular-value basis.

Exercise 6.1 (Ridge and lasso with duplicated predictors; ISLR Ex. 6.5)

Let $n = p = 2$, $x_{11} = x_{12} = a$, $x_{21} = x_{22} = -a$ with $a \neq 0$, and $y_1 + y_2 = 0$, so that the intercept is 0 and is omitted. Let $\lambda > 0$.
(a) Write the ridge problem and show that its solution has $\hat\beta_1 = \hat\beta_2$.
(b) Write the lasso problem and show that its solution is in general not unique; describe all solutions.

Solution. With $s = \beta_1 + \beta_2$ the residuals are $y_1 - as$ and $y_2 + as = -(y_1 - as)$, so $\mathrm{RSS} = 2(y_1 - as)^2$.

(a) Ridge minimizes $R(\beta) = 2(y_1 - a(\beta_1 + \beta_2))^2 + \lambda(\beta_1^2 + \beta_2^2)$.

Step 1. R is strictly convex (a convex quadratic plus the strictly convex $\lambda\|\beta\|^2$) and tends to ∞ as $\|\beta\| \to \infty$, so it has a unique minimizer.

Step 2. $R(\beta_1, \beta_2) = R(\beta_2, \beta_1)$, so swapping the coordinates of the minimizer gives a minimizer; by uniqueness $\hat\beta_1 = \hat\beta_2 = b$. Minimizing $2(y_1 - 2ab)^2 + 2\lambda b^2$ gives $b = 2ay_1/(4a^2 + \lambda)$.

(b) The lasso minimizes $L(\beta) = 2(y_1 - a(\beta_1 + \beta_2))^2 + \lambda(|\beta_1| + |\beta_2|)$.

Step 1. $|\beta_1| + |\beta_2| \geq |s|$, with equality iff $\beta_1\beta_2 \geq 0$. Hence $L(\beta) \geq q(s) := 2a^2(s - m)^2 + \lambda|s|$ with $m = y_1/a$, and equality holds iff β_1, β_2 have the same sign (or one is 0).

Step 2. q is strictly convex with unique minimizer $\hat s = \mathrm{sign}(m)\big(|m| - \lambda/(4a^2)\big)_+$ (the soft-thresholding computation of the orthonormal-design theorem, with $4a^2$ in place of 2).

Step 3. For every β, $L(\beta) \geq q(s) \geq q(\hat s)$, and $L(\beta) = q(\hat s)$ iff $\beta_1 + \beta_2 = \hat s$ and $\beta_1\beta_2 \geq 0$. So the lasso solutions are

$$\{(\beta_1, \beta_2) : \ \beta_1 + \beta_2 = \hat s, \ \beta_1\beta_2 \geq 0\},$$

the segment from $(\hat s, 0)$ to $(0, \hat s)$: infinitely many if $\hat s \neq 0$, only $(0, 0)$ if $|y_1| \leq \lambda/(4|a|)$. ■

Exercise 6.2 (Bayesian view of ridge and lasso; ISLR Ex. 6.7)

Let $y_i = \sum_{j=1}^p x_{ij}\beta_j + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ (centered data, σ^2 known).
(a) Write the likelihood.
(b) With independent double-exponential priors $p(\beta_j) = \frac{1}{2b}e^{-|\beta_j|/b}$, write the posterior and show that its mode is the lasso estimate for some λ.
(c) With independent $N(0, c)$ priors, show that the posterior is normal and that its mode and its mean both equal the ridge estimate for some λ.

Solution.

(a) $L(\beta) = (2\pi\sigma^2)^{-n/2}\exp\{-\|y - X\beta\|^2/(2\sigma^2)\}$.

(b) The posterior density is proportional to $L(\beta)\prod_j p(\beta_j)$:

$$p(\beta \mid y) \propto \exp\Big\{-\frac{1}{2\sigma^2}\Big(\|y - X\beta\|^2 + \frac{2\sigma^2}{b}\|\beta\|_1\Big)\Big\}.$$

As $u \mapsto e^{-u/(2\sigma^2)}$ is strictly decreasing, the mode minimizes $\mathrm{RSS} + \lambda\|\beta\|_1$, $\lambda = 2\sigma^2/b$.

(c)

Step 1. Here $p(\beta \mid y) \propto \exp\{-Q(\beta)/(2\sigma^2)\}$ with $Q(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$, $\lambda = \sigma^2/c$.

Step 2. With $A = X^\top X + \lambda I \succ 0$ and $m = A^{-1}X^\top y$, using $m^\top A = y^\top X$,

$$Q(\beta) = \beta^\top A\beta - 2\beta^\top X^\top y + y^\top y = (\beta - m)^\top A(\beta - m) + y^\top y - m^\top Am.$$

Step 3. So $p(\beta \mid y) \propto \exp\{-\frac{1}{2}(\beta - m)^\top(\sigma^2 A^{-1})^{-1}(\beta - m)\}$, the $N(m, \sigma^2 A^{-1})$ density, whose mean and mode are $m = (X^\top X + \lambda I)^{-1}X^\top y$, the ridge estimate. ■

Exercise 6.3 (Ridge in the singular-value basis)

Let $X = \sum_{j=1}^r d_j u_j v_j^\top$ be a singular value decomposition ($d_j > 0$, $r = \mathrm{rank} X$, orthonormal u_j and v_j), $\lambda > 0$, $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1}X^\top y$ and $H_\lambda = X(X^\top X + \lambda I)^{-1}X^\top$.

(a) Show that $\hat\beta_\lambda = \sum_{j=1}^r \dfrac{d_j\, u_j^\top y}{d_j^2 + \lambda}\, v_j$.

(b) Show that $\|\hat\beta_\lambda\|$ is nonincreasing in λ, strictly decreasing unless $X^\top y = 0$, and tends to 0.

(c) Show that $\mathrm{df}(\lambda) = \mathrm{tr} H_\lambda = \sum_{j=1}^r d_j^2/(d_j^2 + \lambda)$, decreasing from r (as $\lambda \to 0$) to 0.

Solution [Supplement].

(a) Extend $v_1, \dots, v_r$ to an orthonormal basis $v_1, \dots, v_p$ of $\mathbb{R}^p$. Then $(X^\top X + \lambda I)v_j = (d_j^2 + \lambda)v_j$ for $j \le r$ and $= \lambda v_j$ for $j > r$, so $(X^\top X + \lambda I)^{-1}v_j = v_j/(d_j^2 + \lambda)$ for $j \le r$. Since $X^\top y = \sum_{j \le r} d_j(u_j^\top y)v_j$, the formula follows.

(b) By orthonormality,

$$\|\hat\beta_\lambda\|^2 = \sum_{j=1}^{r} \frac{d_j^2 (u_j^\top y)^2}{(d_j^2 + \lambda)^2},$$

each term nonincreasing in λ and strictly decreasing when $u_j^\top y \neq 0$. Since $X^\top y = \sum_j d_j(u_j^\top y)v_j$, all $u_j^\top y = 0$ iff $X^\top y = 0$. Each term tends to 0 as $\lambda \to \infty$.

(c) $H_\lambda = X\hat\beta_\lambda$ as a map of y, i.e. $H_\lambda y = \sum_j \frac{d_j^2}{d_j^2+\lambda}(u_j^\top y)u_j$, so $H_\lambda = \sum_j \frac{d_j^2}{d_j^2+\lambda} u_j u_j^\top$ and $\mathrm{tr} H_\lambda = \sum_j \frac{d_j^2}{d_j^2+\lambda}$ (as $\mathrm{tr} u_j u_j^\top = 1$). Each summand decreases from 1 to 0.

The coordinate along u_j is shrunk by $d_j^2/(d_j^2 + \lambda)$, most for low-variance principal components (PCR keeps or drops them). ■

Chapter 7. Nonlinear Regression

Lecture 6-4 (polynomials, step functions, basis functions, regression splines, smoothing splines, local regression, GAM)

1 Basis-Function Regression

1.1 Polynomial regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i.$$

Fitted by OLS on $X_1 = X$, $X_2 = X^2$, …; the target is the fitted function

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4,$$

with pointwise interval $\hat{f}(x_0) \pm t_{1-\alpha/2,\mathrm{df}}\,\mathrm{SE}[\hat{f}(x_0)]$, where $\mathrm{SE}[\hat{f}(x_0)]^2 = \ell_0^\top \widehat{\mathrm{Var}}(\hat{\beta})\ell_0$ with $\ell_0 = (1, x_0, \ldots, x_0^4)^\top$. d is rarely above 3 or 4, or chosen by CV. Auto data: $d = 1$ underfits, $d = 2$ fits well, $d = 14$ is wild.

Limitation. Polynomials are global: distant data affect the local fit and the tails become erratic for large d.

1.2 Step functions

Cut the range of X at $c_1 < \cdots < c_K$:

$$C_0(X) = \mathbb{I}(X < c_1),\ C_1(X) = \mathbb{I}(c_1 \le X < c_2),\ \ldots,\ C_K(X) = \mathbb{I}(c_K \le X),$$

so $C_0 + \cdots + C_K = 1$, and fit $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$ (C_0 is dropped because of the intercept). β_0 is the mean for $X < c_1$, β_k the difference of bin k. Wage on age with bins < 35, 35–50, 50–65, ≥ 65: the first bin misses the increasing trend. Results depend on the breakpoints.

1.3 Basis functions

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

with fixed b_k: $b_j(x) = x^j$, or $b_j(x) = \mathbb{I}(c_j \le x < c_{j+1})$; least squares, SEs and F-tests apply. Other bases: Fourier, wavelets, splines.

2 Regression Splines

2.1 Piecewise polynomials

Separate low-degree polynomials between **knots**; with one knot c,

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \varepsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \varepsilon_i, & x_i \ge c. \end{cases}$$

Unconstrained pieces jump at the knot; splines impose the maximum continuity.

2.2 Linear splines

Definition: linear spline

A linear spline with knots $\xi_1 < \cdots < \xi_K$ is piecewise linear and continuous at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \varepsilon_i, \qquad b_1(x) = x, \quad b_{k+1}(x) = (x - \xi_k)_+,$$

with $(x - \xi)_+ = x - \xi$ if $x > \xi$ and 0 otherwise.

$(x - \xi_k)_+$ adds a slope change β_{k+1} at ξ_k.

(a) basis functions (b) linear spline



Figure 7: (a) Truncated power basis, knots ξ_1, ξ_2. (b) A linear-spline fit.

2.3 Cubic splines

Definition: cubic spline and truncated power basis

A cubic spline with knots $\xi_1, \ldots, \xi_K$ is piecewise cubic with continuous derivatives up to order 2 at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i,$$

$$b_1(x) = x, \; b_2(x) = x^2, \; b_3(x) = x^3, \; b_{k+3}(x) = (x - \xi_k)_+^3, \; k = 1, \ldots, K.$$

Theorem: degrees of freedom of a cubic spline

The space of cubic splines with K knots has dimension $K + 4$, and the truncated power basis (with the constant) is a basis.

Proof [Supplement]. $K + 1$ unconstrained cubic pieces have $4(K + 1)$ parameters; continuity of the function, first and second derivatives at each of the K knots imposes $3K$ linear constraints, leaving $4(K + 1) - 3K = K + 4$. Each $(x - \xi_k)_+^3$ is twice continuously differentiable (its first two derivatives vanish at ξ_k) and its third derivative jumps by 6 at ξ_k, so the $K+4$ functions are linearly independent splines. ■

2.4 Natural cubic splines

Splines have high variance at the boundary. A **natural cubic spline** is also *linear* beyond the boundary knots: 4 more constraints (second and third derivatives zero at both ends), so K knots give K degrees of freedom (including the intercept) and narrower boundary bands.

2.5 Number and locations of knots

Place knots at quantiles of X; choose K by CV. Wage data: a degree-15 polynomial is wild at the boundaries, a 15-df natural cubic spline is well behaved.

3 Smoothing Splines

Motivation. Make every data point a knot and control wiggliness by a roughness penalty.

Definition: smoothing spline

Among all twice-differentiable functions g,

$$\min_g \underbrace{\sum_{i=1}^{n}(y_i - g(x_i))^2}_{\text{RSS}} + \lambda \underbrace{\int g''(t)^2\,dt}_{\text{roughness penalty}} .$$

$\lambda = 0$: interpolation; $\lambda \to \infty$: a linear g (zero penalty). λ by CV.

Theorem: the minimizer is a natural cubic spline

The minimizer $\hat{g}_\lambda$ is a natural cubic spline with knots at the unique values of $x_1, \ldots, x_n$ — a shrunken version of the interpolating natural spline. Consequently $\hat{g}_\lambda = (\hat{g}(x_1), \ldots, \hat{g}(x_n))^\top = S_\lambda y$ for an $n \times n$ smoother matrix S_λ.

Idea [Supplement]. The natural cubic spline $\tilde{g}$ interpolating $g(x_i)$ has the same RSS and $\int \tilde{g}''^2 \le \int g''^2$, with equality only if $g = \tilde{g}$ (Exercise 7.3). These splines form an n-dimensional space $g(x) = \sum_j N_j(x)\theta_j$, so the problem becomes $\min_\theta \|y - N\theta\|^2 + \lambda\theta^\top \Omega_N \theta$ with $(\Omega_N)_{jk} = \int N_j'' N_k''$, solved by $\hat{\theta} = (N^\top N + \lambda\Omega_N)^{-1} N^\top y$, a generalized ridge regression. Hence $S_\lambda = N(N^\top N + \lambda\Omega_N)^{-1} N^\top$. ■

Effective degrees of freedom $\mathrm{df}_\lambda = \mathrm{tr}(S_\lambda)$ decrease from n ($\lambda = 0$) to 2 ($\lambda \to \infty$). LOOCV costs one fit: $\mathrm{CV}(\lambda) = \frac{1}{n}\sum_i \{(y_i - \hat{g}_\lambda(x_i))/(1 - \{S_\lambda\}_{ii})\}^2$ (Ch. 3). Wage data: about 6.8 df (by LOOCV) is nearly indistinguishable from 16 df.

4 Local Regression

Weighted linear fits in a sliding window.

Algorithm: local regression at a target point x_0

1. Gather the fraction $s = k/n$ of training points closest to x_0 (the **span**).
2. Assign weights $K_{i0} = K(x_i, x_0)$: zero outside the neighborhood, largest for the closest points (e.g. tricube weights).
3. Fit a weighted least-squares regression $\min_{\beta_0,\beta_1} \sum_i K_{i0}(y_i - \beta_0 - \beta_1 x_i)^2$.
4. The fitted value is $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Choices: weights, local degree (constant, linear, quadratic), and span s (small: wiggly; large: global; by CV). Local linear fits remove the boundary bias of local constant averages. Curse of dimensionality beyond three or four predictors.

5 Generalized Additive Models

Nonlinear in each variable, additive across variables:

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i.$$

With natural splines a GAM is one least-squares fit; terms may also be smoothing splines, local regression, linear or step functions.

Algorithm: backfitting (for smoothing-spline or local-regression terms)

Initialize $\hat{\beta}_0 = \bar{y}$, $\hat{f}_j = 0$. Repeat until convergence: for $j = 1, \ldots, p$, smooth the partial residuals $r_i = y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik})$ against x_{ij} to update $\hat{f}_j$, then center $\hat{f}_j$ to have mean zero.

Example: Wage data

$\texttt{wage} = \beta_0 + f_1(\texttt{year}) + f_2(\texttt{age}) + f_3(\texttt{education}) + \varepsilon$, natural splines for year and age, a constant per education level. With the others fixed, wage rises slightly with year, rises then falls with age, and increases with education.

Pros. Nonlinear f_j without manual transformations; each effect can still be examined with the others fixed; df summarize each f_j. **Cons.** Additivity misses interactions unless added, e.g. $f_{jk}(X_j, X_k)$.

Summary

- Basis expansions (polynomials, steps, splines) stay within least squares; cubic splines have $K + 4$ df, natural cubic splines K.
- Smoothing splines are natural cubic splines with knots at the data; $\mathrm{df}_\lambda = \mathrm{tr} S_\lambda$.
- Local regression: weighted LS in a window; GAMs: additive nonlinear terms (backfitting).

6 Selected exercises

Motivation. ISLR Ex. 7.1 and 7.2, and the least-curvature property behind smoothing splines.

Exercise 7.1 (The truncated power basis gives a cubic spline; ISLR Ex. 7.1)

Let $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$.
(a) Find cubics $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$ and $f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$ with $f = f_1$ on $x \leq \xi$ and $f = f_2$ on $x > \xi$.
(b) Show that $f_1(\xi) = f_2(\xi)$, $f_1'(\xi) = f_2'(\xi)$ and $f_1''(\xi) = f_2''(\xi)$, so f is a cubic spline with knot ξ.

Solution.

(a) For $x \leq \xi$ the truncated term vanishes: $a_1 = \beta_0$, $b_1 = \beta_1$, $c_1 = \beta_2$, $d_1 = \beta_3$. For $x > \xi$, $(x - \xi)^3 = x^3 - 3\xi x^2 + 3\xi^2 x - \xi^3$, so

$$a_2 = \beta_0 - \beta_4 \xi^3, \quad b_2 = \beta_1 + 3\beta_4 \xi^2, \quad c_2 = \beta_2 - 3\beta_4 \xi, \quad d_2 = \beta_3 + \beta_4.$$

(b) As polynomials, $f_2 - f_1 = \beta_4(x-\xi)^3$, with derivatives $3\beta_4(x-\xi)^2$ and $6\beta_4(x-\xi)$. All three vanish at $x = \xi$. (The third derivative jumps by $6\beta_4$.) ■

Exercise 7.2 (Which penalty, which limit; ISLR Ex. 7.2)

Let $x_1 < \cdots < x_n$ be distinct points in $[a, b]$, $n \geq 3$, and

$$\hat{g} = \arg\min_g \sum_{i=1}^{n}(y_i - g(x_i))^2 + \lambda \int_a^b \left(g^{(m)}(x)\right)^2 dx$$

over functions with continuous mth derivative ($g^{(0)} = g$). Interpreting $\lambda = \infty$ as the constraint that the penalty vanishes, describe $\hat{g}$ for $\lambda = \infty$ and $m = 0, 1, 2, 3$, and for $\lambda = 0$, $m = 3$.

Solution.

Step 1. A continuous nonnegative function with zero integral vanishes identically, so $\int_a^b (g^{(m)})^2 = 0$ iff $g^{(m)} \equiv 0$ on $[a, b]$. For $m = 0$ this means $g \equiv 0$; for $m \geq 1$, integrating m times, it means that g is a polynomial of degree at most $m - 1$.

Step 2. With $\lambda = \infty$, $\hat{g}$ minimizes RSS over that class:

(a) $m = 0$: $\hat{g} \equiv 0$.

(b) $m = 1$: constants; $\sum_i (y_i - c)^2$ is minimized at $c = \bar{y}$, so $\hat{g} \equiv \bar{y}$.

(c) $m = 2$: lines; $\hat{g}$ is the least-squares line.

(d) $m = 3$: quadratics; $\hat{g}$ is the least-squares quadratic, unique because three or more distinct x_i make the design $(1, x_i, x_i^2)$ of full column rank.

Step 3. With $\lambda = 0$, $m = 3$, every smooth g with $g(x_i) = y_i$ (e.g. the interpolating polynomial of degree $n - 1$) attains the minimum RSS $= 0$: $\hat{g}$ interpolates and is not unique. ■

Exercise 7.3 (The natural cubic spline has least curvature)

Let $a < x_1 < \cdots < x_n < b$ with $n \geq 2$, and let $\tilde{g}$ be a natural cubic spline with knots $x_1, \ldots, x_n$ (linear on $[a, x_1]$ and $[x_n, b]$). Let g be any twice continuously differentiable function on $[a, b]$ with $g(x_i) = \tilde{g}(x_i)$ for all i. Show that $\int_a^b g''^2 \geq \int_a^b \tilde{g}''^2$, with equality iff $g = \tilde{g}$. Deduce that the smoothing-spline minimizer is a natural cubic spline with knots at the x_i.

Solution [Supplement]. Let $h = g - \tilde{g}$; it is twice continuously differentiable and $h(x_i) = 0$ for all i.

Step 1. $\tilde{g}''$ is continuous and piecewise linear, hence absolutely continuous with derivative $\tilde{g}'''$ almost everywhere, where $\tilde{g}'''$ equals a constant c_j on (x_j, x_{j+1}) and 0 on (a, x_1) and (x_n, b). Integration by parts gives

$$\int_a^b \tilde{g}''h'' = \left[\tilde{g}''h'\right]_a^b - \int_a^b \tilde{g}'''h'.$$

The boundary term vanishes because $\tilde{g}$ is linear near a and b, so $\tilde{g}''(a) = \tilde{g}''(b) = 0$.

Step 2. $\int_a^b \tilde{g}'''h' = \sum_{j=1}^{n-1} c_j \int_{x_j}^{x_{j+1}} h' = \sum_{j=1}^{n-1} c_j\{h(x_{j+1}) - h(x_j)\} = 0$. Hence $\int \tilde{g}''h'' = 0$.

Step 3. Therefore

$$\int_a^b g''^2 = \int_a^b (\tilde{g}'' + h'')^2 = \int_a^b \tilde{g}''^2 + \int_a^b h''^2 \;\geq\; \int_a^b \tilde{g}''^2.$$

Step 4. Equality forces $\int h''^2 = 0$, so $h'' \equiv 0$ (continuity) and h is linear. A linear function vanishing at $n \geq 2$ points is 0, so $g = \tilde{g}$.

Step 5 (consequence). For any candidate g, the natural cubic spline $\tilde{g}$ interpolating $g(x_i)$ has the same RSS and, by Steps 3–4, a strictly smaller penalty unless $g = \tilde{g}$. ■

Chapter 8. Classification Errors and the Bayes Classifier

Lecture 7-1 (error rate, Bayes optimal classifier, KNN, bias–variance, confusion matrix, ROC/AUC, CV for classification)

1 The Classification Problem

Accuracy is measured by the **error rate**

$$\text{Error rate} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(y_i \neq \hat{y}_i),$$

Goals: a classifier $C(X)$ with labels in $\mathcal{C}$, the uncertainty of each classification, the roles of the predictors.

2 The Bayes Optimal Classifier

Motivation. Assigning each x to its most probable class gives a lower bound on every method's error.

Classes $1, \ldots, K$; risk $R(C) = \Pr(C(X) \neq Y)$.

Theorem: Bayes classifier

The risk is minimized by

$$C^*(x) = \arg\max_k \Pr(Y = k \mid X = x),$$

assigning each x to the class with the largest conditional probability. Its risk $R(C^*) = 1 - \mathbb{E}\big[\max_k \Pr(Y = k \mid X)\big]$ is the **Bayes error rate**, the lowest error achievable by any classifier.

Proof [Supplement]. $R(C) = 1 - \Pr(C(X) = Y) = 1 - \mathbb{E}\big[\Pr(Y = C(X) \mid X)\big]$. For each x, $\Pr(Y = C(x) \mid X = x) \leq \max_k \Pr(Y = k \mid X = x)$, with equality when $C(x) = C^*(x)$. Taking expectations, $R(C) \geq R(C^*)$. ■

The Bayes error is positive whenever the classes overlap. **Bayes decision boundary**: $\{x : \Pr(Y = 1 \mid x) = 0.5\}$.

3 Approximating the Bayes Classifier

Test error of $\hat{C}$:

$$\text{Err}_{\text{Te}} = \text{Ave}_{i \in \text{Te}}\, \mathbb{I}\{y_i \neq \hat{C}(x_i)\}.$$

Either *estimate the conditional probabilities* and plug in (logistic regression, GAMs, LDA/QDA, KNN), or *model* $C(x)$ *directly* (support-vector machines).

3.1 K-nearest neighbors

$$\hat{\Pr}(Y = k \mid X = x) = \frac{1}{K}\sum_{x_i \in \mathcal{N}_K(x)} \mathbb{I}(y_i = k),$$

and predict the majority class. **Smaller K, more complex**: $K = 1$ is overly flexible (training error 0), $K = 100$ nearly linear, $K = 10$ close to the Bayes boundary.

3.2 Bias–variance in classification

As $1/K$ grows, training error declines; test error is U-shaped.

4 Other Measures for Classification

Motivation. With rare classes the overall error rate can look excellent for a useless classifier.

4.1 Confusion matrix

Default data (LDA fit, $n = 10{,}000$):

		True default status		
		No	Yes	Total
Predicted	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

- Overall error: $(23 + 252)/10000 = 2.75\%$ — but always predicting "No" would already give $333/10000 = 3.33\%$.
- **False positive rate**: fraction of negatives classified positive, $23/9667 = 0.2\%$.
- **False negative rate**: fraction of positives classified negative, $252/333 = 75.7\%$.

Sensitivity (recall, TPR) $= 1 - \text{FNR} = 81/333 = 24.3\%$; specificity $= 1 - \text{FPR} = 99.8\%$: unacceptable for catching defaulters.

4.2 Thresholds

Threshold 0.5 mimics the Bayes rule; lowering it (0.2, 0.1) trades fewer false negatives for more false positives. With costs c_{FN}, c_{FP}: predict "Yes" iff $\hat{p}\, c_{FN} > (1 - \hat{p}) c_{FP}$, i.e. $\hat{p} > c_{FP}/(c_{FP} + c_{FN})$ (Exercise 8.1).

4.3 ROC curve and AUC

ROC curve: TPR against FPR as the threshold moves from 1 to 0. **AUC**: area under it; 0.5 is random guessing, 1 perfect separation.

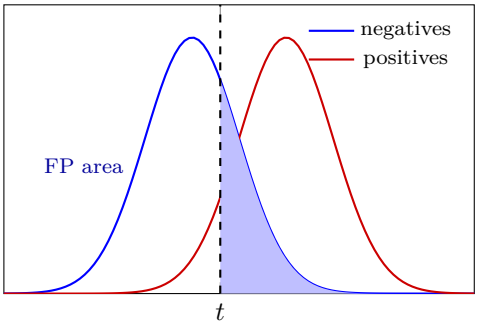
Theorem: probabilistic meaning of AUC

For a score $s(x)$, $\text{AUC} = \Pr\left(s(X^+) > s(X^-)\right)$ for independent draws X^+ from the positive and X^- from the negative class (ties counted 1/2).

Proof [Supplement]. With threshold t, $\text{TPR}(t) = \Pr(s^+ > t)$ and $\text{FPR}(t) = \Pr(s^- > t)$. Then $\text{AUC} = \int \text{TPR}\, d\text{FPR} = \int \Pr(s^+ > t)\, dF_-(t) = \Pr(s^+ > s^-)$. ■

AUC depends only on the ranking: invariant to monotone transformations and class proportions.

(a) score distributions



(b) ROC curve

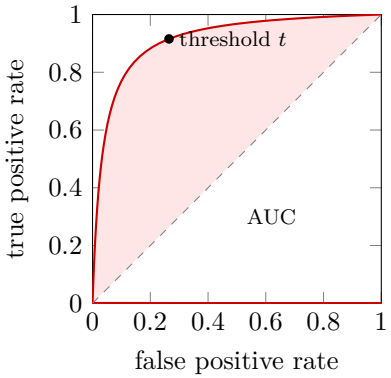


Figure 8: (a) Score distributions; shaded: FPR at threshold t. (b) Moving t traces the ROC curve.

5 Cross-Validation for Classification

With parts $C_1, \ldots, C_K$ of sizes n_k,

$$\mathrm{CV}_K = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{Err}_k, \qquad \mathrm{Err}_k = \frac{1}{n_k} \sum_{i \in C_k} \mathbb{I}(y_i \neq \hat{y}_i),$$

$\hat{y}_i$ fitted without part k. For polynomial logistic regression and KNN on simulated data, 10-fold CV may underestimate the test error level but its minimum is near the right complexity.

Summary

- The Bayes classifier $\arg\max_k \Pr(Y = k \mid x)$ attains the lowest error; KNN votes locally.
- Report FPR, FNR, sensitivity, specificity; choose thresholds by costs; $\mathrm{AUC} = \Pr(s^+ > s^-)$.

6 Selected exercises

Motivation. The cost-weighted threshold and the two-Gaussian Bayes error used in Ch. 10.

Exercise 8.1 (The cost-sensitive Bayes rule)

Let $Y \in \{0, 1\}$, $\eta(x) = \Pr(Y = 1 \mid X = x)$, and let a classifier C incur cost $c_{FN} > 0$ when $C(X) = 0$, $Y = 1$ and $c_{FP} > 0$ when $C(X) = 1$, $Y = 0$. Show that the expected cost

$$R(C) = c_{FN} \Pr(C(X) = 0, Y = 1) + c_{FP} \Pr(C(X) = 1, Y = 0)$$

is minimized by $C^*(x) = \mathbb{I}\{\eta(x) > c_{FP}/(c_{FP} + c_{FN})\}$, which is the Bayes classifier when $c_{FP} = c_{FN}$.

Solution [Supplement].

Step 1. Conditioning on X,

$$R(C) = \mathbb{E}\big[c_{FN}\, \eta(X) \mathbb{I}\{C(X) = 0\} + c_{FP}\{1 - \eta(X)\}\mathbb{I}\{C(X) = 1\}\big].$$

Step 2. For each x the integrand is $c_{FN}\eta(x)$ if $C(x) = 0$ and $c_{FP}\{1 - \eta(x)\}$ if $C(x) = 1$, so it is at least $\min\{c_{FN}\eta(x), c_{FP}(1 - \eta(x))\}$, the value attained by C^*: $C^*(x) = 1$ exactly when $c_{FP}(1 - \eta) < c_{FN}\eta$, i.e. $\eta > c_{FP}/(c_{FP} + c_{FN})$ (at equality both choices cost the same).

Step 3. Taking expectations, $R(C) \geq R(C^*)$. Equal costs: threshold $1/2$; $c_{FN} = 9c_{FP}$: threshold 0.1 (Default data, Ch. 10). ■

Exercise 8.2 (Bayes error of two Gaussian classes)

Let $\pi_1 = \pi_2 = \frac{1}{2}$ and $X \mid Y = k \sim N(\mu_k, \Sigma)$ in $\mathbb{R}^p$, $\Sigma \succ 0$, $\mu_1 \neq \mu_2$, and $\Delta^2 = (\mu_2 - \mu_1)^\top \Sigma^{-1}(\mu_2 - \mu_1)$ (Mahalanobis distance).
(a) Show that the Bayes classifier assigns class 2 iff $w^\top x > w^\top(\mu_1 + \mu_2)/2$, with $w = \Sigma^{-1}(\mu_2 - \mu_1)$.
(b) Show that the Bayes error is $\Phi(-\Delta/2)$. Check the value $\Phi(-1.5) \approx 6.7\%$ for $p = 1$, $\mu_{1,2} = \mp 1.5$, $\sigma = 1$.

Solution [Supplement].

(a) With equal priors the Bayes rule picks the larger density. The normalizing constants are equal, so

$$\log \frac{f_2(x)}{f_1(x)} = \frac{1}{2}\{(x - \mu_1)^\top \Sigma^{-1}(x - \mu_1) - (x - \mu_2)^\top \Sigma^{-1}(x - \mu_2)\} = w^\top x - \frac{1}{2} w^\top(\mu_1 + \mu_2),$$

where the terms $x^\top \Sigma^{-1} x$ cancel and $\mu_2^\top \Sigma^{-1}\mu_2 - \mu_1^\top \Sigma^{-1}\mu_1 = (\mu_1 + \mu_2)^\top \Sigma^{-1}(\mu_2 - \mu_1)$ by symmetry of Σ^{-1}. Class 2 wins iff this is positive.

(b)

Step 1. Put $c = w^\top(\mu_1 + \mu_2)/2$. Under class 1, $w^\top X \sim N(w^\top \mu_1, w^\top \Sigma w)$ with $w^\top \Sigma w = \Delta^2$, and $c - w^\top \mu_1 = \frac{1}{2} w^\top(\mu_2 - \mu_1) = \Delta^2/2$. Hence $\Pr(w^\top X > c \mid Y = 1) = 1 - \Phi(\Delta/2) = \Phi(-\Delta/2)$.

Step 2. Under class 2, $w^\top \mu_2 - c = \Delta^2/2$, so $\Pr(w^\top X \leq c \mid Y = 2) = \Phi(-\Delta/2)$.

Step 3. The Bayes error is $\frac{1}{2}\Phi(-\Delta/2) + \frac{1}{2}\Phi(-\Delta/2) = \Phi(-\Delta/2)$. For $p = 1$, $\Delta = |\mu_2 - \mu_1|/\sigma = 3$, giving $\Phi(-1.5) = 0.0668$. ■

Chapter 9. Discriminative Methods: Logistic Regression

Lecture 7-2 (discriminative vs generative, logit model, GLM, maximum likelihood, confounding, extensions, multinomial regression)

1 Discriminative Methods versus Generative Models

Discriminative: model $\Pr(Y = c_k \mid X = x)$ directly	Generative: model (X, Y), then derive $\Pr(Y = c_k \mid X = x)$
K-nearest neighbors, logistic regression, classification trees, ensembles (bagging, random forest, boosting), support vector machines	linear and quadratic discriminant analysis, naive Bayes, Gaussian mixture models

Examples: tumours benign/malignant; transactions legitimate/fraudulent; news topics. Often we estimate $P(Y = k \mid X)$ and classify to the most probable class.

2 Why Not Linear Regression?

With $Y \in \{0, 1\}$ and $\hat{Y} = \beta_0 + \beta X$: $\mathbb{E}[Y \mid X] = P(Y = 1 \mid X) \in [0, 1]$, but a linear fit is unbounded. (For two classes the classification rule equals LDA's, but the probabilities are poor.)

Coding $Y = 1$ (stroke), 2 (drug overdose), 3 (epileptic seizure) imposes a meaningless ordering and spacing; different codings give different fits.

3 The Logistic Model

For $Y \in \{c_1, c_2\}$, the **logit** model:

$$\log \frac{\Pr(Y = c_1 \mid X)}{\Pr(Y = c_2 \mid X)} = \beta_0 + \beta_1 X,$$

$$\Pr(Y = c_1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}, \quad \Pr(Y = c_2 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X}},$$

an S-shaped curve in $[0, 1]$. β_1: change in the **log odds** per unit of X; e^{β_1} multiplies the odds; $\partial p / \partial X = \beta_1 p(1 - p)$.

3.1 Logistic regression as a GLM

A GLM: Y from an exponential family, linear predictor $\eta = X^\top \beta$, link $g(\mathbb{E}[Y]) = \eta$. Here $Y \sim$ Bernoulli(p) with the canonical logit link, the natural parameter of $p^y (1 - p)^{1-y} = \exp\{y \log \frac{p}{1-p} + \log(1 - p)\}$.

4 Estimation by Maximum Likelihood

Motivation. No closed form: maximize the likelihood; standard errors come from its curvature.

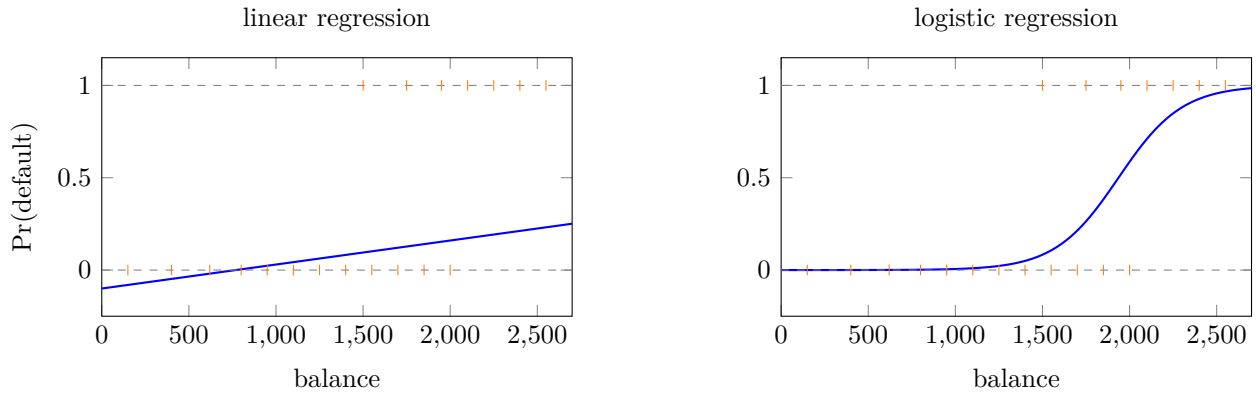


Figure 9: Default data (schematic). The linear fit leaves $[0,1]$; the logistic fit $\hat{p} = e^{-10.65+0.0055X}/(1+e^{-10.65+0.0055X})$ does not.

$$\mathcal{L}(\beta_0,\beta) = \prod_{i:y_i=1} p(x_i) \prod_{j:y_j=0} \{1-p(x_j)\}, \qquad \ell(\beta_0,\beta_1) = \sum_i \Big[y_i(\beta_0+\beta_1 x_i) - \log\left(1+e^{\beta_0+\beta_1 x_i}\right)\Big].$$

Theorem: score equations and concavity

With $p_i = p(x_i)$ and design rows $\tilde{x}_i = (1, x_i)^\top$,

$$\nabla \ell = \sum_i (y_i - p_i)\tilde{x}_i = \tilde{X}^\top (y-p), \qquad \nabla^2 \ell = -\tilde{X}^\top W \tilde{X},\ W = \mathrm{diag}\{p_i(1-p_i)\}.$$

ℓ is concave, so the MLE solves $\tilde{X}^\top(y-p)=0$ and is found by Newton–Raphson, equivalently **iteratively reweighted least squares**: $\beta^{\text{new}} = (\tilde{X}^\top W \tilde{X})^{-1}\tilde{X}^\top W z$ with working response $z = \tilde{X}\beta + W^{-1}(y-p)$.

Proof [Supplement]. $\frac{d}{d\eta}\log(1+e^\eta) = \frac{e^\eta}{1+e^\eta} = p$ and $\frac{dp}{d\eta} = p(1-p)$; differentiate termwise. The Hessian is negative semidefinite. A Newton step $\beta + (\tilde{X}^\top W \tilde{X})^{-1}\tilde{X}^\top(y-p)$ rearranges into the weighted least-squares form. ■

$\widehat{\mathrm{Var}}(\hat{\beta}) = (\tilde{X}^\top \hat{W} \tilde{X})^{-1}$ (inverse Fisher information) gives SEs and z-statistics. Under perfect separation the MLE does not exist (Exercise 9.1; Ch. 10).

5 Logistic Regression on the Default Data

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

One unit of balance adds 0.0055 to the log odds; $H_0 : \beta_1 = 0$ is rejected ($p < 0.0001$).

Predictions.

$$\hat{p}(1000) = \frac{e^{-10.6513+0.0055\times 1000}}{1+e^{-10.6513+0.0055\times 1000}} = 0.006, \qquad \hat{p}(2000) = \frac{e^{-10.6513+0.0055\times 2000}}{1+e^{-10.6513+0.0055\times 2000}} = 0.586.$$

A qualitative predictor. With student status: intercept -3.5041, student[Yes] 0.4049 ($p = 0.0004$), so $\Pr(\text{default} \mid \text{student}) = e^{-3.0992}/(1+e^{-3.0992}) = 0.0431$ and $\Pr(\text{default} \mid \text{non-student}) = e^{-3.5041}/(1+e^{-3.5041}) = 0.0292$.

6 Multiple Logistic Regression and Confounding

$$\log \frac{p(X)}{1-p(X)} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p, \qquad p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}.$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.8690	0.4923	−22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income (thousands)	0.0030	0.0082	0.37	0.7115
student[Yes]	−0.6468	0.2362	−2.74	0.0062

The student coefficient is now *negative*. **Confounding**: students carry higher balances, so their marginal default rate is higher, but at fixed balance they default *less*.

7 Extensions of Logistic Regression

- **Polynomial logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d$. E.g. `wage` > 250 on a degree-4 polynomial in age.
- **Piecewise-constant logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i)$ with step functions of age.
- **GAM for classification**: $\log \frac{P(X)}{1-P(X)} = \beta_0 + f_1(X_1) + \cdots + f_p(X_p)$; for the Wage data, β_1 `year` $+ f_2($`age`$) + f_3($`education`$)$ with a smoothing spline (5 df) in age and a step function for education.
- **Penalized logistic regression**: minimize $-\log \mathcal{L}(\beta_0, \beta) + \lambda\|\beta\|_2^2$ (ridge) or $+\lambda\|\beta\|_1$ (lasso), generally $+\lambda\|\beta\|_q^q$; the penalty also gives finite estimates under separation.
- **Multinomial logistic regression** (below).

8 More Than Two Classes: Multinomial Regression

Baseline class K; for $k = 1, \ldots, K-1$,

$$\log \frac{\Pr(Y=k \mid X=x)}{\Pr(Y=K \mid X=x)} = \beta_{k0} + \beta_{k1} x_1 + \cdots + \beta_{kp} x_p.$$

Theorem: implied probabilities

$$\Pr(Y=k \mid X=x) = \frac{e^{\beta_{k0}+\beta_k^\top x}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}} \quad (k<K),$$

$$\Pr(Y=K \mid X=x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}},$$

and the log odds between any two classes k, k' is linear: $\log \frac{\Pr(Y=k|x)}{\Pr(Y=k'|x)} = (\beta_{k0}-\beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$.

Proof [Supplement]. Write $\pi_k = \Pr(Y=k \mid x) = \pi_K e^{\eta_k}$ for $k < K$; the probabilities sum to 1, so $\pi_K(1+\sum_l e^{\eta_l}) = 1$. The difference of two log odds against K gives the pairwise log odds. ■

The baseline affects only the parametrization. Softmax: $\Pr(Y=k \mid x) = e^{\eta_k} / \sum_{l=1}^{K} e^{\eta_l}$, identifiable up to a common shift (Exercise 9.4).

Summary

- Logistic regression: linear log odds (Bernoulli GLM, logit link), fitted by IRLS.
- Multiple regression reveals confounding; multinomial regression has linear pairwise log odds.

9 Selected exercises

Motivation. Separation, a binary predictor, and ISLR Ex. 4.6 and 4.12.

Exercise 9.1 (No MLE under complete separation)

Let $\ell(\beta) = \sum_i [y_i \tilde{x}_i^\top \beta - \log(1 + e^{\tilde{x}_i^\top \beta})]$ with $y_i \in \{0, 1\}$, and suppose some b satisfies $\tilde{x}_i^\top b > 0$ whenever $y_i = 1$ and $\tilde{x}_i^\top b < 0$ whenever $y_i = 0$.
(a) Show that $\ell(\beta) < 0$ for every β and $\ell(tb) \to 0$ as $t \to \infty$; hence no maximizer exists.
(b) Show that for $\lambda > 0$ the ridge-penalized objective $\ell(\beta) - \lambda\|\beta\|^2$ has a unique maximizer.

Solution [Supplement].

(a) With $p_i = e^{\tilde{x}_i^\top \beta}/(1+e^{\tilde{x}_i^\top \beta}) \in (0, 1)$, each term of ℓ equals $\log p_i$ if $y_i = 1$ and $\log(1-p_i)$ if $y_i = 0$, the log of a number in $(0, 1)$; so $\ell(\beta) < 0$. Along $\beta = tb$: if $y_i = 1$, $\tilde{x}_i^\top b > 0$ and $p_i \to 1$; if $y_i = 0$, $\tilde{x}_i^\top b < 0$ and $p_i \to 0$. Every term tends to 0, so $\sup_\beta \ell = 0$, which is never attained.

(b) $F(\beta) = \ell(\beta) - \lambda\|\beta\|^2$ is continuous, $F \le -\lambda\|\beta\|^2 \to -\infty$ as $\|\beta\| \to \infty$, so the set $\{F \ge F(0)\}$ is closed and bounded and F attains its maximum there. ℓ is concave (its Hessian is $-\tilde{X}^\top W \tilde{X}$) and $-\lambda\|\beta\|^2$ is strictly concave, so F is strictly concave and the maximizer is unique. ■

Exercise 9.2 (Logistic regression on one binary predictor)

Let $x_i \in \{0, 1\}$; group x has n_x observations of which s_x have $y_i = 1$, with $0 < s_x < n_x$, and $\hat{p}_x = s_x/n_x$. Fit $\log \frac{p}{1-p} = \beta_0 + \beta_1 x$.
(a) Show that $\hat{\beta}_0 = \log \frac{\hat{p}_0}{1-\hat{p}_0}$ and $\hat{\beta}_1 = \log \frac{\hat{p}_1/(1-\hat{p}_1)}{\hat{p}_0/(1-\hat{p}_0)}$ (the log odds ratio), so the fitted probabilities are the group proportions.
(b) Show that whenever the model has an intercept, $\sum_i \hat{p}_i = \sum_i y_i$.

Solution [Supplement].

(a)

Step 1. The score equations $\tilde{X}^\top (y - p) = 0$ read $\sum_i (y_i - p_i) = 0$ and $\sum_i x_i (y_i - p_i) = 0$.

Step 2. In group x all p_i equal $p(x) = \sigma(\beta_0 + \beta_1 x)$, $\sigma(\eta) = e^\eta/(1 + e^\eta)$. The second equation gives $s_1 - n_1 p(1) = 0$; subtracting it from the first gives $s_0 - n_0 p(0) = 0$. So $p(0) = \hat{p}_0$, $p(1) = \hat{p}_1$.

Step 3. Inverting σ yields $\hat{\beta}_0 = \text{logit}(\hat{p}_0)$ and $\hat{\beta}_0 + \hat{\beta}_1 = \text{logit}(\hat{p}_1)$. Both groups are present, so $\tilde{X}$ has full column rank and $W \succ 0$; the Hessian $-\tilde{X}^\top W \tilde{X}$ is negative definite, ℓ is strictly concave, and this solution is the unique MLE.

(b) The intercept's score equation is $\sum_i (y_i - \hat{p}_i) = 0$.

Default data: 0.0292 and 0.0431 are the non-student and student default rates; $e^{0.4049} \approx 1.50$ is their odds ratio. ■

Exercise 9.3 (Reading a logistic fit; ISLR Ex. 4.6)

For students, X_1 = hours studied, X_2 = undergraduate GPA and Y = receiving an A, a logistic fit gives $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.
(a) Estimate the probability of an A for a student who studies 40 hours and has GPA 3.5.
(b) How many hours would this student need for a 50% chance?

Solution.

(a) The linear predictor is $-6+0.05(40)+3.5 = -0.5$, so $\hat{p} = e^{-0.5}/(1+e^{-0.5}) = 1/(1+e^{0.5}) \approx 0.378$.

(b) $\hat{p} = 0.5$ iff the log odds are 0: $-6 + 0.05h + 3.5 = 0$, i.e. $h = 50$ hours. ■

Exercise 9.4 (Logistic versus softmax coding; ISLR Ex. 4.12)

To classify $X \in \mathbb{R}$ into apples and oranges, you fit $\Pr(\text{orange} \mid x) = e^{\beta_0+\beta_1 x}/(1 + e^{\beta_0+\beta_1 x})$; a friend fits the softmax model $\Pr(\text{orange} \mid x) = e^{\alpha_{o0}+\alpha_{o1}x}/(e^{\alpha_{o0}+\alpha_{o1}x} + e^{\alpha_{a0}+\alpha_{a1}x})$, both by maximum likelihood on the same data.
(a) Give the log odds of orange versus apple in both models.
(b) If $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = -1$, what are the friend's estimates?
(c) If the friend gets $\hat{\alpha}_{o0} = 1.2$, $\hat{\alpha}_{o1} = -2$, $\hat{\alpha}_{a0} = 3$, $\hat{\alpha}_{a1} = 0.6$, what are yours?
(d) How often will the two predicted labels agree on 2000 test points?

Solution.

(a) Yours: $\beta_0 + \beta_1 x$. The friend's: $\log \frac{e^{\alpha_{o0}+\alpha_{o1}x}}{e^{\alpha_{a0}+\alpha_{a1}x}} = (\alpha_{o0} - \alpha_{a0}) + (\alpha_{o1} - \alpha_{a1})x$.

Key fact. By (a), the softmax probabilities equal the logistic ones with $\beta = \alpha_o - \alpha_a$, so the friend's likelihood is $L_{\text{soft}}(\alpha) = L_{\log}(\alpha_o - \alpha_a)$. As α ranges over $\mathbb{R}^4$, $\alpha_o - \alpha_a$ ranges over all of $\mathbb{R}^2$; hence $\hat{\alpha}$ maximizes L_{soft} iff $\hat{\alpha}_o - \hat{\alpha}_a$ maximizes $L_{\log}$.

(b) Only differences are determined: $\hat{\alpha}_{o0} - \hat{\alpha}_{a0} = 2$ and $\hat{\alpha}_{o1} - \hat{\alpha}_{a1} = -1$, e.g. $\hat{\alpha}_a = (0, 0)$, $\hat{\alpha}_o = (2, -1)$, or any common shift of both.

(c) $\hat{\beta}_0 = 1.2 - 3 = -1.8$ and $\hat{\beta}_1 = -2 - 0.6 = -2.6$.

(d) Always (100%): the two fits give identical probabilities at every x, hence identical labels. ■

Chapter 10. Generative Models for Classification

Lecture 7-3 (Bayes theorem, LDA, QDA, naive Bayes, logistic regression vs LDA, comparison of classifiers)

1 Discriminant Analysis as Generative Modeling

Generative models describe

$$\Pr(X, Y) = \Pr(X \mid Y)\Pr(Y)$$

and classify by Bayes' rule. Discriminant analysis:

(1) model X in each class, (2) flip with Bayes' theorem. Gaussian classes:

$$X \mid Y = c_k \sim N(\mu_k, \Sigma) \;\; \text{(LDA)}, \qquad X \mid Y = c_k \sim N(\mu_k, \Sigma_k) \;\; \text{(QDA)};$$

1.1 Bayes' theorem for classification

With class densities f_k and priors $\pi_k = \Pr(Y = k)$,

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)} \;\propto\; \pi_k f_k(x).$$

$\hat{C}(x) = \arg\max_k \pi_k f_k(x)$ (the Bayes classifier of Ch. 8, estimated). Favoring a class moves the boundary away from it.

1.2 Why discriminant analysis?

Logistic regression is unstable for well-separated classes (Exercise 9.1); LDA is not, and it is also stabler for small n with near-normal classes. It suits $K > 2$ and gives low-dimensional views.

2 Linear Discriminant Analysis

Motivation. Model each class density as normal and apply Bayes' rule.

2.1 One predictor

$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\{-\frac{1}{2}(\frac{x-\mu_k}{\sigma_k})^2\}$ with $\sigma_1 = \cdots = \sigma_K = \sigma$:

$$p_k(x) = \frac{\pi_k \exp\{-\frac{1}{2\sigma^2}(x-\mu_k)^2\}}{\sum_{l=1}^{K} \pi_l \exp\{-\frac{1}{2\sigma^2}(x-\mu_l)^2\}}.$$

Theorem: discriminant functions

Classifying to the largest $p_k(x)$ is equivalent to classifying to the largest

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k,$$

a *linear* function of x. With $K = 2$ and $\pi_1 = \pi_2 = 0.5$ the decision boundary is $x = (\mu_1 + \mu_2)/2$.

Proof [Supplement]. The denominator does not depend on k. Taking logs, $\log(\pi_k f_k(x)) = \log \pi_k - \frac{(x-\mu_k)^2}{2\sigma^2} + c = \log \pi_k - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + c$, and $-x^2/(2\sigma^2) + c$ is common to all classes. For two classes with equal priors, $\delta_1(x) = \delta_2(x)$ gives $x(\mu_1 - \mu_2) = (\mu_1^2 - \mu_2^2)/2$. ■

Example: simulation

$\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, $\sigma^2 = 1$: Bayes boundary $x = 0$, Bayes error $\Phi(-1.5) \approx 6.7\%$ (Exercise 8.2). With 20 observations per class the LDA boundary is close to 0 and its error slightly above.

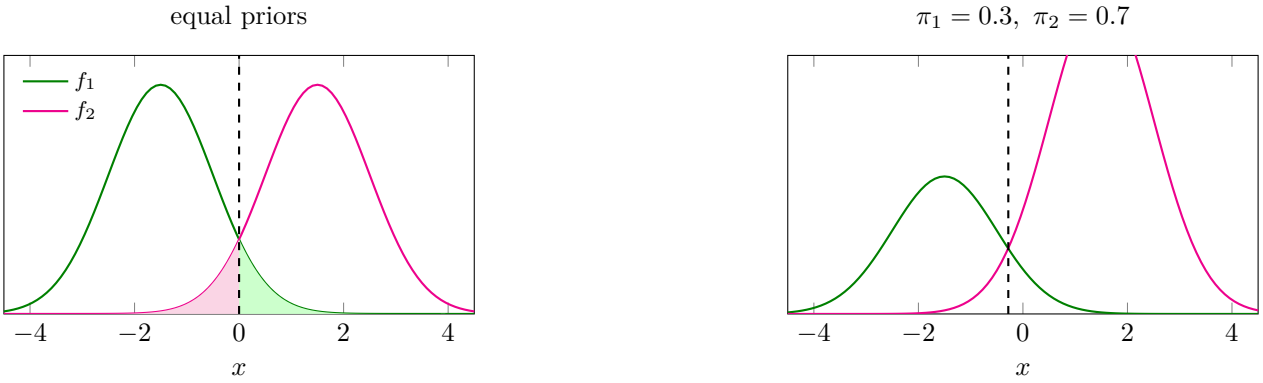


Figure 10: $\pi_k f_k$ for $\mu_{1,2} = \mp 1.5$, $\sigma = 1$. Right: the boundary moves to $\frac{\mu_1+\mu_2}{2} + \frac{\sigma^2 \log(\pi_1/\pi_2)}{\mu_2-\mu_1} \approx -0.28$.

2.2 Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2 \quad \text{(pooled variance)},$$

the full-likelihood MLEs except for the divisor $n - K$ (Exercise 10.2).

2.3 Several predictors

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\Big\{ -\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big\},$$

with $\Sigma_k = \Sigma$ for all k.

Theorem: LDA in p dimensions

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k, \qquad \hat{C}(x) = \arg\max_k \delta_k(x).$$

The boundary between classes k and k' is the hyperplane

$$\Big\{x :\ x^\top \Sigma^{-1}(\mu_k - \mu_{k'}) + \log \frac{\pi_k}{\pi_{k'}} - \frac{1}{2}(\mu_k + \mu_{k'})^\top \Sigma^{-1}(\mu_k - \mu_{k'}) = 0\Big\},$$

with normal vector $\Sigma^{-1}(\mu_k - \mu_{k'})$, generally *not* in the direction of $\mu_k - \mu_{k'}$.

Proof [Supplement]. As in one dimension: $\log(\pi_k f_k(x)) = \log \pi_k - \frac{1}{2}(x-\mu_k)^\top \Sigma^{-1}(x-\mu_k) + c$, and the quadratic term $-\frac{1}{2}x^\top \Sigma^{-1} x$ is shared by all classes because Σ is common. Set $\delta_k = \delta_{k'}$. ■

Iris data (4 variables, 3 species, 50 each): LDA misclassifies 3 of 150 training samples. **Default data**: training error 2.75%, FPR 0.2%, FNR 75.7% (Ch. 8).

2.4 Adjusting the errors

LDA uses threshold 0.5 on $\Pr(\text{Yes} \mid \text{balance}, \text{student})$. Lowering it to 0.2 cuts the Default FNR from 75.7% to about 41% with a slightly higher overall error; likewise for logistic regression.

3 Quadratic Discriminant Analysis

Class-specific Σ_k:

$$\delta_k(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log\pi_k = x^\top W_k x + x^\top w_k + b_k,$$

with $W_k = -\frac{1}{2}\Sigma_k^{-1}$, $w_k = \Sigma_k^{-1}\mu_k$ and $b_k = -\frac{1}{2}\mu_k^\top\Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| + \log\pi_k$: a quadratic form. The boundary between k and k' is the quadric

$$\{x:\ x^\top(W_k - W_{k'})x + x^\top(w_k - w_{k'}) + (b_k - b_{k'}) = 0\}.$$

Parameter count. Covariance parameters: LDA $p(p+1)/2$, QDA $Kp(p+1)/2$ (1275 versus 3825 for $p = 50$, $K = 3$). QDA suits clearly different covariances and enough data; LDA suits similar covariances or few data (lower variance, possibly higher bias).

4 Naive Bayes

Inputs independent *within each class*:

$$f(x_1,\ldots,x_p \mid y = c_k) = f(x_1 \mid c_k) f(x_2 \mid c_k)\cdots f(x_p \mid c_k).$$

Gaussian naive Bayes takes each Σ_k diagonal:

$$\delta_k(x) \ \propto\ \log\Big[\pi_k \prod_{j=1}^{p} f_{kj}(x_j)\Big] = -\frac{1}{2}\sum_{j=1}^{p}\Big[\frac{(x_j-\mu_{kj})^2}{\sigma_{kj}^2} + \log\sigma_{kj}^2\Big] + \log\pi_k.$$

Only one-dimensional densities are needed (Gaussian, kernel, or frequency tables). It often works when n is not large relative to p: the bias is outweighed by the variance reduction, and only the *ranking* of posteriors matters.

5 Comparison of Classification Methods

5.1 Logistic regression versus LDA

LDA log posterior odds:

$$\log\frac{\Pr(y=c_k \mid x)}{\Pr(y=c_K \mid x)} = x^\top\Sigma^{-1}(\mu_k - \mu_K) + \log\frac{\pi_k}{\pi_K} - \frac{1}{2}(\mu_k+\mu_K)^\top\Sigma^{-1}(\mu_k-\mu_K) = \alpha_{k0} + \alpha_k^\top x,$$

and logistic regression has linear logits $\beta_{k0} + \beta_k^\top x$ by construction. They are different estimators:

- **Origin of linearity.** For LDA a consequence of Gaussian classes with common covariance.

- **Common component.** In $\Pr(X = x, Y = c_k) = \Pr(X = x)\Pr(Y = c_k \mid x)$ both have the multinomial-logit second factor $\exp(\beta_{k0} + \beta_k^\top x)/(1 + \sum_{k'} \exp(\beta_{k'0} + \beta_{k'}^\top x))$.
- **Different assumptions on $\Pr(X)$.** Logistic: arbitrary; LDA: the mixture $\sum_k \pi_k \phi(x; \mu_k, \Sigma)$.
- **Estimation.** Logistic maximizes the *conditional* likelihood (marginal left nonparametric, mass $1/n$ per point); LDA the *full* likelihood of $\phi(x; \mu_k, \Sigma)\pi_k$.

Remarks. Under Gaussian classes logistic regression may lose about 30% asymptotic efficiency in error rate (Efron, 1975). LDA uses all points to estimate Σ (outlier sensitive); logistic regression down-weights points far from the boundary (tiny $p_i(1 - p_i)$).

5.2 QDA versus LDA, logistic regression and KNN

Linear true boundary: LDA and logistic regression; moderately nonlinear: QDA; complicated: KNN with K by CV, given enough data.

Method	Boundary	Strengths / weaknesses
logistic regression	linear (or by basis expansion)	few assumptions; probabilities; unstable under separation
LDA	linear	efficient under Gaussian classes; stable with small n; outlier sensitive
QDA	quadratic	flexible covariances; many parameters
naive Bayes	depends on f_{kj}	scales to large p; independence assumption
KNN	arbitrary	nonparametric; needs much data; no interpretation

Summary

- Classify by $\arg\max_k \pi_k f_k(x)$: LDA linear, QDA quadratic, naive Bayes trades bias for variance.
- LDA and logistic regression share the logit form but differ in $\Pr(X)$ and in full versus conditional likelihood.

6 Selected exercises

Motivation. ISLR Ex. 4.3 and 4.7, and the MLE property of the LDA estimates.

Exercise 10.1 (QDA in one dimension; ISLR Ex. 4.3)

Let $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$ with priors π_k, $k = 1, \ldots, K$.
(a) Show that the Bayes classifier assigns x to the largest

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log \sigma_k + \log \pi_k.$$

(b) Show that if $\sigma_k \neq \sigma_l$, then $\delta_k - \delta_l$ is a genuine quadratic, the two classes are separated by at most two points, and the class with the larger variance wins in both tails; so the Bayes classifier is not linear.

Solution.

(a) The Bayes classifier maximizes $\pi_k f_k(x)$, equivalently its logarithm

$$\log \pi_k - \tfrac{1}{2}\log(2\pi) - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}.$$

Drop the common $-\frac{1}{2}\log(2\pi)$ and expand; unlike LDA, $-x^2/(2\sigma_k^2)$ depends on k.

(b) $\delta_k(x) - \delta_l(x) = Ax^2 + Bx + C$ with $A = \frac{1}{2\sigma_l^2} - \frac{1}{2\sigma_k^2} \neq 0$. A nonzero quadratic has at most two real roots, so the line splits into at most three intervals, alternately assigned to k and l. If $\sigma_k > \sigma_l$, then $A > 0$ and $\delta_k - \delta_l \to +\infty$ as $|x| \to \infty$: class k wins on both tails, which no single threshold (linear rule) can produce. ■

Exercise 10.2 (LDA estimates are maximum-likelihood estimates)

Observe (x_i, y_i), $i = 1, \ldots, n$, with $x_i \in \mathbb{R}$, $y_i \in \{1, \ldots, K\}$, class sizes $n_k \geq 1$, and full likelihood $\prod_i \pi_{y_i}\phi(x_i; \mu_{y_i}, \sigma^2)$. Assume $W = \sum_k \sum_{i:y_i=k}(x_i - \bar{x}_k)^2 > 0$, $\bar{x}_k$ the class means.
(a) Show that the MLE is $\hat{\pi}_k = n_k/n$, $\hat{\mu}_k = \bar{x}_k$, $\hat{\sigma}^2_{\mathrm{ML}} = W/n$.
(b) Show that $\mathbb{E}W = (n-K)\sigma^2$, so the pooled $\hat{\sigma}^2 = W/(n-K)$ is the unbiased rescaling.

Solution [Supplement].

(a) The log-likelihood separates:

$$\sum_k n_k \log \pi_k \;-\; \frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_k \sum_{i:y_i=k}(x_i - \mu_k)^2.$$

Step 1 (π). For $\pi_k > 0$, $\sum_k \pi_k = 1$, Jensen's inequality for the concave log gives

$$\sum_k n_k \log \pi_k - \sum_k n_k \log \frac{n_k}{n} = n \sum_k \frac{n_k}{n} \log \frac{n\pi_k}{n_k} \leq n \log \sum_k \pi_k = 0,$$

with equality iff $n\pi_k/n_k$ is constant, i.e. $\pi_k = n_k/n$.

Step 2 (μ). For any σ^2 the last term is maximized by minimizing each $\sum_{i:y_i=k}(x_i - \mu_k)^2 = \sum_{i:y_i=k}(x_i - \bar{x}_k)^2 + n_k(\bar{x}_k - \mu_k)^2$, i.e. at $\mu_k = \bar{x}_k$; the minimum value of the double sum is W.

Step 3 (σ^2). $s \mapsto -\frac{n}{2}\log s - \frac{W}{2s}$ has derivative $(W - ns)/(2s^2)$, positive for $s < W/n$ and negative beyond, so the maximum is at W/n.

(b) Within class k the x_i are i.i.d. $N(\mu_k, \sigma^2)$, so $\mathbb{E}\sum_{i:y_i=k}(x_i - \bar{x}_k)^2 = (n_k - 1)\sigma^2$; summing over k gives $(n-K)\sigma^2$. ■

Exercise 10.3 (Predicting a dividend; ISLR Ex. 4.7)

Among companies that issued a dividend (80% of them) last year's percentage profit X has mean 10; among the others, mean 0; both have variance 36, and X is normal within each group. Find the probability that a company with $X = 4$ issues a dividend.

Solution. By Bayes' theorem with $f_k(x) \propto \exp\{-(x-\mu_k)^2/72\}$ (the common factor $1/\sqrt{72\pi}$ cancels),

$$\Pr(\text{Yes} \mid X = 4) = \frac{0.8\, e^{-(4-10)^2/72}}{0.8\, e^{-(4-10)^2/72} + 0.2\, e^{-(4-0)^2/72}} = \frac{0.8\, e^{-1/2}}{0.8\, e^{-1/2} + 0.2\, e^{-2/9}} \approx \frac{0.4852}{0.6454} \approx 0.752.$$

$X = 4$ is closer to 0, but the prior 0.8 tips the posterior. ■

Chapter 11. Tree-Based Methods and Ensembles

Lecture 8 (regression and classification trees, pruning, bagging, OOB, random forests, boosting, AdaBoost, GBM, XGBoost, LightGBM, CatBoost, importance, SHAP)

1 Decision Trees

Trees *stratify* the predictor space into simple regions, for regression and classification. They are interpretable but not very accurate; bagging, random forests and boosting combine many trees, improving accuracy at the price of interpretability.

2 Regression Trees

2.1 Example: Hitters data

Log salary from `Years` (in the major leagues) and `Hits` (previous year); two internal nodes, three leaves:

$$R_1 = \{X : \texttt{Years} < 4.5\}, \quad R_2 = \{X : \texttt{Years} \geq 4.5,\ \texttt{Hits} < 117.5\},$$
$$R_3 = \{X : \texttt{Years} \geq 4.5,\ \texttt{Hits} \geq 117.5\},$$

each leaf predicting its mean response. Regions are **terminal nodes**, split points **internal nodes**.

Interpretation. Years matters most; Hits matters only for players with five or more years: an interaction a linear model would need to be told about.

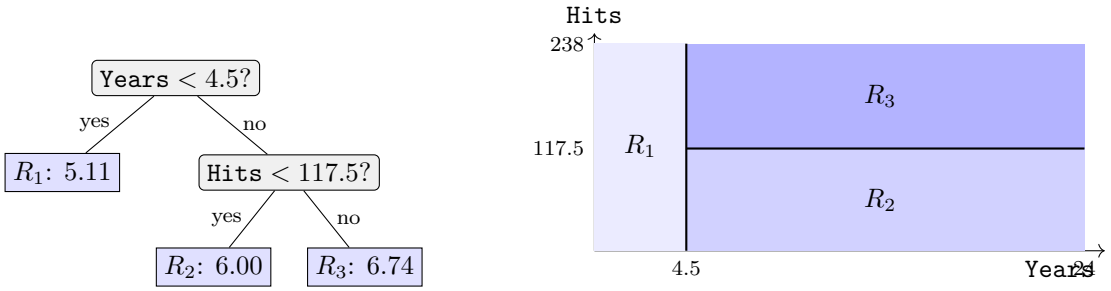


Figure 11: The Hitters tree (mean log salaries) and its partition.

2.2 Building a tree

1. Divide the predictor space into J distinct, non-overlapping regions $R_1, \ldots, R_J$.
2. For every observation in R_j, predict the mean response of the training observations in R_j.

With boxes as regions, minimize

$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2,$$

$\hat{y}_{R_j}$ the mean response in box j.

2.3 Recursive binary splitting

All partitions are infeasible; use a **top-down, greedy** search (best split *at each step*, no look-ahead).

Algorithm: recursive binary splitting

Select the predictor X_j and cutpoint s that minimize

$$\sum_{i:x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i:x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2,$$

$$R_1(j,s) = \{X : X_j < s\}, \; R_2(j,s) = \{X : X_j \geq s\}.$$

Then repeat within one of the resulting regions (not the whole space), and so on, until a stopping criterion is reached (e.g. no region contains more than five observations).

The search over s is fast (sort, running sums).

2.4 Pruning

A large tree overfits. Stopping when a split gains little is short-sighted (a worthless split may precede a very good one); instead grow a large T_0 and prune.

Definition: cost-complexity (weakest-link) pruning

For each $\alpha \geq 0$ there is a subtree $T \subset T_0$ minimizing

$$\sum_{m=1}^{|T|} \sum_{i:x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|,$$

where $|T|$ is the number of terminal nodes, R_m the rectangle of the mth terminal node and $\hat{y}_{R_m}$ its mean.

$\alpha = 0$ gives T_0; as α grows, branches are pruned in a nested way: collapsing node t (branch T_t) pays off once $\alpha \geq \{R(t) - R(T_t)\}/(|T_t| - 1)$, the weakest link first.

Algorithm: building a regression tree

1. Grow a large tree by recursive binary splitting, stopping only when each terminal node has fewer than some minimum number of observations.
2. Apply cost-complexity pruning to obtain the sequence of best subtrees as a function of α.
3. Choose α by K-fold CV: for each k, repeat steps 1–2 on all but the kth fold and evaluate the MSE on the kth fold as a function of α; average and pick $\hat{\alpha}$.
4. Return the subtree from step 2 corresponding to $\hat{\alpha}$.

Hitters (132 training, 131 test): six-fold CV picks three leaves, close to the test-error minimum.

3 Classification Trees

Predict the most common class in the region (proportions as probabilities); split by an impurity instead of RSS.

Definition: node impurity measures

With $\hat{p}_{mk}$ the proportion of training observations in region m from class k:

$$E = 1 - \max_k \hat{p}_{mk} \text{ (classification error)}, \qquad G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk}) \text{ (Gini)},$$

$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk} \text{ (cross-entropy)}.$$

The error is not sensitive enough for growing. Gini (sum of Bernoulli variances) is small when all $\hat{p}_{mk}$ are near 0 or 1: **node purity**. Cross-entropy is numerically similar.

Example: why Gini is more sensitive than the error rate [Supplement]

400 cases per class, split into $(300, 100)$, $(100, 300)$ or into $(200, 400)$, $(200, 0)$: both have error 0.25, but Gini $\frac{1}{2}(0.375) + \frac{1}{2}(0.375) = 0.375$ versus $\frac{3}{4}(0.444) + \frac{1}{4}(0) = 0.333$ prefers the pure node.

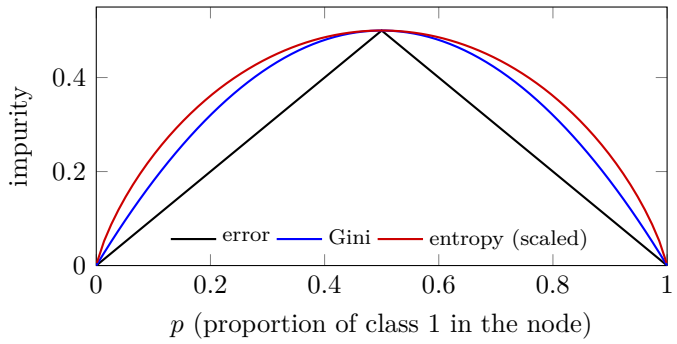


Figure 12: Two-class node impurity (entropy scaled to 0.5 at $p = \frac{1}{2}$).

Gini and entropy are strictly concave, so any split weakly decreases the weighted impurity (Jensen); the piecewise-linear error often shows no gain. For pruning, the error rate is preferred when accuracy is the goal.

Heart data: binary HD for 303 chest-pain patients, 13 predictors; CV gives six leaves. Qualitative predictors are split directly by levels.

3.1 Trees versus linear models

Linear: $f(X) = \beta_0 + \sum_j \beta_j X_j$; tree: $f(X) = \sum_{m=1}^{M} c_m \mathbb{I}\{X \in R_m\}$. Near-linear truth favors the linear model, box-like truth the tree. A circular boundary $X_1^2 + X_2^2 = 4.6$ is approximated by a staircase of boxes.

3.2 Advantages and disadvantages of trees

Easy to explain and display, close to human decisions, no dummy variables needed. But less accurate and non-robust: a small data change can change the tree a lot.

4 Bagging

4.1 Idea

Averaging reduces variance (σ^2/n for a mean). Bootstrap aggregation fits $\hat{f}^{*b}$ on B bootstrapped training sets and averages:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{*b}(x).$$

Classification: **majority vote** (or averaged probabilities). Trees are grown deep, unpruned (low bias, high variance).

Theorem: variance of an average of identically distributed predictors

If $\hat{f}^{*1}, \ldots, \hat{f}^{*B}$ at a point x each have variance σ^2 and pairwise correlation ρ, then

$$\text{Var}\Big(\frac{1}{B}\sum_{b} \hat{f}^{*b}(x)\Big) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Proof [Supplement]. $\text{Var}(\sum_b \hat{f}^{*b}) = B\sigma^2 + B(B-1)\rho\sigma^2$; divide by B^2. ■

The $\rho\sigma^2$ term remains as $B \to \infty$: overlapping bootstrap samples correlate the trees. Large B does not overfit.

4.2 Out-of-bag error estimation

Theorem: each bagged tree uses about two thirds of the observations

The probability that a given observation is *not* in a bootstrap sample is $(1 - 1/n)^n \to e^{-1} \approx 0.368$.

Proof [Supplement]. Each of the n independent draws misses it with probability $1 - 1/n$; and $n \log(1 - 1/n) \to -1$. ■

Each observation is **out-of-bag** (OOB) for about $B/3$ trees; averaging their predictions gives a valid test-error estimate, for large B essentially the LOOCV error without refitting. Heart data: OOB errors track test errors; bagging and random forests beat a single tree.

5 Random Forests

Trees on bootstrap samples, but each split may use only one of m predictors sampled afresh from the p; typically $m \approx \sqrt{p}$ (classification) or $p/3$ (regression). This **decorrelates** the trees: a strong predictor no longer dominates every top split, lowering ρ. $m = p$ is bagging.

Gene-expression data: 4,718 genes, 349 patients, 15 classes; on the 500 highest-variance genes, $m < p$ slightly beats bagging; a single tree errs 45.7%. No overfitting as B grows.

6 Boosting

6.1 Boosting regression trees

Trees are grown **sequentially**, each using the previous ones; no bootstrap.

Algorithm: boosting for regression trees

1. Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all i.
2. For $b = 1, \dots, B$: fit a tree $\hat{f}^b$ with d splits ($d+1$ leaves) to the data (X, r); update $\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x)$ and the residuals $r_i \leftarrow r_i - \lambda \hat{f}^b(x_i)$.
3. Output $\hat{f}(x) = \sum_{b=1}^{B} \lambda \hat{f}^b(x)$.

Boosting **learns slowly**: small trees on the residuals, further slowed by λ.

Three tuning parameters.

- B: boosting can (slowly) overfit; choose by CV.
- λ: learning rate, typically 0.01 or 0.001; small λ needs large B.
- d: interaction depth; $d = 1$ (stumps) gives an additive model.

Gene-expression data: boosted stumps ($\lambda = 0.01$) slightly beat depth-2 boosting and random forests.

6.2 AdaBoost

Algorithm: AdaBoost, $y \in \{-1, +1\}$

Start with weights $D_1(i) = 1/n$. For $b = 1, \dots, B$:
(i) train a weak learner h_b using weights D_b;
(ii) compute its weighted error $\epsilon_b = \sum_i D_b(i) \mathbb{I}\{h_b(x_i) \neq y_i\}$;
(iii) set $\alpha_b = \frac{1}{2} \log \frac{1-\epsilon_b}{\epsilon_b}$;
(iv) update $D_{b+1}(i) = D_b(i) \exp\{-\alpha_b y_i h_b(x_i)\}/Z_b$, Z_b a normalizer. Output $H(x) = \text{sign}\big(\sum_{b=1}^{B} \alpha_b h_b(x)\big)$.

Misclassified observations gain weight; accurate learners get a larger vote.

Theorem: AdaBoost minimizes the exponential loss stagewise

With $F_{b-1} = \sum_{s<b} \alpha_s h_s$, the pair (α_b, h_b) minimizes $\sum_i \exp\{-y_i(F_{b-1}(x_i) + \alpha h(x_i))\}$.

Proof [Supplement]. With $D_b(i) \propto e^{-y_i F_{b-1}(x_i)}$, the objective is proportional to $e^{-\alpha}(1-\epsilon) + e^{\alpha}\epsilon$, which for $\alpha > 0$ is increasing in the weighted error ϵ of h (so h_b minimizes it), and setting the α-derivative to zero gives $e^{2\alpha} = (1-\epsilon)/\epsilon$. The new weights are $e^{-y_i F_b(x_i)}$, i.e. the update rule. ■

Toy example: three stumps with $\epsilon = 0.30, 0.21, 0.14$ and votes $0.42, 0.65, 0.92$.

6.3 Gradient boosting machines

Gradient descent in function space for a differentiable loss L:

(i) at step b compute pseudo-residuals $r_{ib} = -\partial L(y_i, F(x_i))/\partial F(x_i)$ at $F = F_{b-1}$;

(ii) fit a tree to them;

(iii) update $F_b = F_{b-1} + \lambda h_b$. For $\frac{1}{2}(y-F)^2$ the pseudo-residual is $y - F$: the algorithm above.

Problem	Loss	Negative gradient
regression	squared $\frac{1}{2}(y-F)^2$; absolute $\lvert y-F\rvert$; Huber	$y-F$; $\text{sign}(y-F)$; clipped residual
classification	Bernoulli deviance $\log(1+e^{-2yF})$; exponential e^{-yF}	$2y/(1+e^{2yF})$; ye^{-yF}

GBM can overfit, has no explicit complexity penalty, and is sequential.

6.4 XGBoost, LightGBM, CatBoost

XGBoost: regularized objective

$$\mathcal{L} = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{b=1}^{B} \Omega(f_b), \qquad \Omega(f_b) = \gamma T_b + \frac{\lambda}{2} \sum_{j=1}^{T_b} w_{bj}^2,$$

T_b leaves with scores w_{bj}. A second-order expansion (gradients g_i, Hessians h_i) gives $w_j^* = -\sum_{i \in I_j} g_i / (\sum_{i \in I_j} h_i + \lambda)$ and a closed-form split gain; plus parallel, cache-aware computation.

LightGBM: histogram-based split finding and leaf-wise growth for large, high-dimensional data.

CatBoost: ordered target statistics for categorical features and *ordered boosting* against target leakage and prediction shift.

Method	Main focus	Mechanism
XGBoost	regularization and second-order optimization	controls tree complexity; uses gradients and Hessians
LightGBM	speed and scalability	histogram-based splitting, leaf-wise growth
CatBoost	categorical data, bias reduction	ordered target statistics and ordered boosting

7 Variable Importance

7.1 Impurity-based importance

$$\mathrm{Imp}(X_j) = \sum_{\text{splits on } X_j} \text{reduction in loss}, \qquad \text{final importance} = \frac{1}{B} \sum_{b=1}^{B} \mathrm{Imp}_b(X_j),$$

with RSS, Gini or cross-entropy. Heart data: Thal, Ca and ChestPain lead. A global, non-causal view.

Limitations: biased toward variables with many possible splits; model-dependent; not comparable across models.

7.2 Permutation importance

Shuffle X_j and measure the error increase:

$$\mathrm{Imp}(X_j) = \mathrm{Error}_{\mathrm{perm}(j)} - \mathrm{Error}_{\mathrm{original}}.$$

Model-agnostic, on OOB or validation data; a correlated substitute can mask importance.

7.3 SHAP values

Shapley additive explanations:

$$f(x) = \phi_0 + \sum_{j=1}^{p} \phi_j, \qquad \phi_j = \sum_{S \subseteq \{1,\ldots,p\} \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \big[v(S \cup \{j\}) - v(S)\big],$$

$v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$, $\phi_0 = \mathbb{E}f(X)$; ϕ_j is the average marginal contribution of X_j over all feature orders. The unique attribution with efficiency ($\sum_j \phi_j = f(x) - \phi_0$), symmetry, dummy and additivity; *local* per observation (TreeSHAP for trees).

Method	Type	Characteristic
tree importance	model-specific	fast but biased
permutation	model-agnostic	based on prediction error
SHAP	model-agnostic	local and consistent explanations

Summary

- Trees: recursive binary splitting, cost-complexity pruning with CV.
- Bagging averages deep bootstrap trees (OOB error); random forests decorrelate them.
- Boosting fits small trees to residuals with shrinkage (B, λ, d); AdaBoost: exponential loss; GBM: negative gradient.

Chapter 12. Support Vector Machines

Lecture 9 (hyperplanes, maximal margin classifier, support vector classifier, kernels, multiclass SVM, SVM vs logistic regression)

1 Separating Hyperplanes

1.1 Motivation

For *linearly separable* classes, find the hyperplane with the *biggest separation*. Otherwise soften "separates" and enlarge the feature space.

1.2 Hyperplanes

A flat affine subspace of dimension $p-1$:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0;$$

With $f(X) = \beta_0 + \beta^\top X$ and $y_i = \pm 1$, $f = 0$ is a **separating hyperplane** if $y_i f(x_i) > 0$ for all i; classify by $\operatorname{sign} f(x)$, with confidence $|f(x)|$.

Theorem: distance to a hyperplane

The signed distance of x to $\{f = 0\}$ is $f(x)/\|\beta\|$; if $\|\beta\| = 1$, $f(x)$ itself is the signed distance.

Proof [Supplement]. β is normal to the hyperplane. For a point x_0 on it, the distance is the length of the projection of $x - x_0$ on $\beta/\|\beta\|$: $\beta^\top(x - x_0)/\|\beta\| = (f(x) - f(x_0))/\|\beta\| = f(x)/\|\beta\|$. ■

2 The Maximal Margin Classifier

The **margin** M is the minimum distance from the observations to the hyperplane; the **optimal separating hyperplane** maximizes it:

$$\max_{\beta_0,\beta} M \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1, \qquad y_i(\beta_0 + \beta^\top x_i) \geq M \quad \forall i.$$

Theorem: equivalent convex problem

The problem is equivalent to

$$\min_{\beta_0,\beta} \frac{1}{2}\|\beta\|^2 \quad \text{subject to} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 \quad \forall i,$$

with margin $M = 1/\|\beta\|$.

Proof [Supplement]. Drop the norm constraint and write the condition as $y_i(\beta_0 + \beta^\top x_i)/\|\beta\| \geq M$ (distances, by the previous theorem). The scale of (β_0, β) is arbitrary, so fix it by $\|\beta\| = 1/M$; then the constraints become $y_i(\beta_0 + \beta^\top x_i) \geq 1$ and maximizing $M = 1/\|\beta\|$ is minimizing $\|\beta\|^2/2$, a convex quadratic program. ■

Only the points on the margin, the **support vectors**, determine the solution (Exercise 12.1).

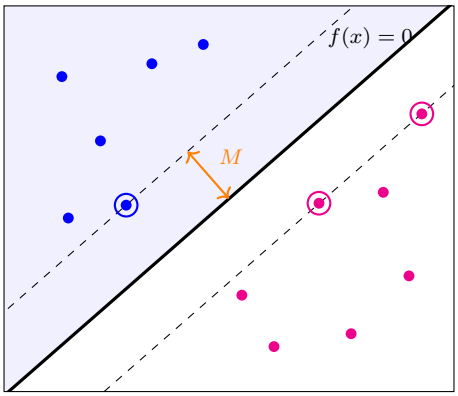


Figure 13: Maximal margin hyperplane and margins; circled: support vectors (schematic).

2.1 Non-separable and noisy data

Data are usually not separable (unless $n < p$), and even then one observation can move the hyperplane dramatically. The support vector classifier maximizes a **soft margin**.

3 The Support Vector Classifier

Definition: soft-margin optimization

$$\max_{\beta_0,\beta,\epsilon} M \quad \text{subject to} \quad \sum_j \beta_j^2 = 1, \quad y_i(\beta_0 + \beta^\top x_i) \geq M(1-\epsilon_i), \quad \epsilon_i \geq 0, \quad \sum_{i=1}^n \epsilon_i \leq C.$$

ϵ_i are **slack variables**; $C \geq 0$ is a tuning parameter.

$\epsilon_i = 0$: correct side of the margin; $0 < \epsilon_i \leq 1$: inside the margin; $\epsilon_i > 1$: misclassified. So at most C training observations are misclassified.

C is a **budget**: larger C, wider margin; $C = 0$ is the maximal margin classifier. Chosen by CV: large C gives many support vectors, low variance, possibly high bias. Only points on or inside the margin matter, so the classifier is robust to far points, unlike LDA.

Derivation: the equivalent penalized form

Equivalently,

$$\min_{\beta_0,\beta,\xi} \frac{1}{2}\|\beta\|^2 + \gamma \sum_i \xi_i \quad \text{s.t.} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 - \xi_i, \ \xi_i \geq 0,$$

where a *large* cost γ corresponds to a *small* budget C (software usually calls γ "C" — the opposite role).

4 Support Vector Machines

4.1 Feature expansion

Enlarge the feature space, e.g. $(X_1, X_2, X_1^2, X_2^2, X_1X_2)$; a linear boundary there,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 = 0,$$

is a conic in the original space. High-degree polynomials get wild fast; kernels are more controlled.

4.2 Inner products and kernels

With $\langle x_i, x_{i'}\rangle = \sum_{j=1}^{p} x_{ij}x_{i'j}$, the linear support vector classifier is

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i\rangle \qquad (n+1 \text{ parameters}),$$

estimated from the $\binom{n}{2}$ pairwise inner products; most $\hat{\alpha}_i$ are zero:

$$f(x) = \hat{\beta}_0 + \sum_{i\in\mathcal{S}} \hat{\alpha}_i \langle x, x_i\rangle,$$

$\mathcal{S} = \{i : \hat{\alpha}_i \neq 0\}$. A **kernel** $K(x, x_i) = \langle \Phi(x), \Phi(x_i)\rangle$ in place of $\langle x, x_i\rangle$ gives the **support vector machine**

$$f(x) = \beta_0 + \sum_{i\in\mathcal{S}} \hat{\alpha}_i K(x, x_i).$$

- **Polynomial kernel** of degree d: $K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij}x_{i'j}\right)^d$.
- **Radial kernel**: $K(x_i, x_{i'}) = \exp\left(-\gamma \sum_{j=1}^{p}(x_{ij} - x_{i'j})^2\right)$, $\gamma > 0$: infinite-dimensional feature space, very local; large γ, more flexible.

Example: the polynomial kernel is an inner product [Supplement]

For $p = 2$, $d = 2$: $(1 + x^\top z)^2 = 1 + 2x_1z_1 + 2x_2z_2 + x_1^2z_1^2 + x_2^2z_2^2 + 2x_1x_2z_1z_2 = \langle \Phi(x), \Phi(z)\rangle$ with $\Phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$.

Kernel versus enlarged feature space. Only the $\binom{n}{2}$ values $K(x_i, x_{i'})$ are needed, while degree-d monomials grow like p^d and the radial kernel has no finite feature set. K is a valid kernel iff every Gram matrix is positive semidefinite (Mercer; Exercise 12.2).

Heart data. In training, radial SVMs with large γ look nearly perfect (overfitting); on the 90 test observations they are worst, while the SV classifier, LDA and the small-γ radial SVM are similar.

5 SVMs With More Than Two Classes

- **One versus all (OVA)**: fit K two-class SVMs $\hat{f}_k(x)$, each class versus the rest; classify x to the class with the largest $\hat{f}_k(x)$.
- **One versus one (OVO)**: fit all $\binom{K}{2}$ pairwise classifiers $\hat{f}_{kl}(x)$; classify x to the class that wins the most pairwise competitions.

For moderate K, use OVO.

6 SVM versus Logistic Regression

Motivation. The SVM is hinge loss plus penalty, and the hinge loss is close to the logistic loss.

Theorem: SVM as loss plus penalty

The support vector classifier with $f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$ solves

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} \underbrace{\max[0,\, 1 - y_i f(x_i)]}_{\text{hinge loss}} + \lambda \sum_{j=1}^{p} \beta_j^2,$$

with large λ corresponding to a large budget C.

Proof [Supplement]. In the penalized form, at the optimum each slack equals $\xi_i = \max[0, 1 - y_i f(x_i)]$ (the smallest feasible value). Substituting gives $\frac{1}{2}\|\beta\|^2 + \gamma \sum_i \max[0, 1 - y_i f(x_i)]$; divide by γ and set $\lambda = 1/(2\gamma)$. ■

The hinge loss vanishes for $y_i f \geq 1$ (hence only support vectors matter); the logistic loss $\log(1 + e^{-y_i f(x_i)})$ is never zero but small for $y_i f \gg 0$.

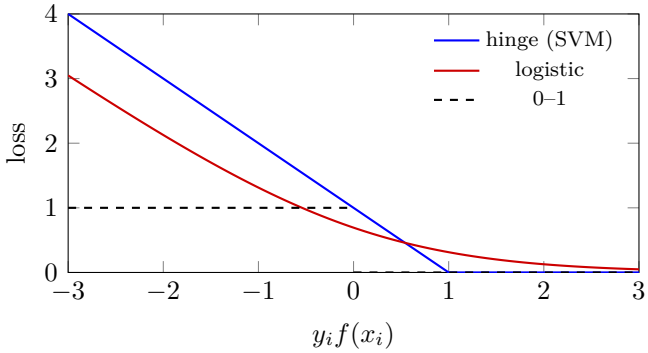


Figure 14: Hinge and logistic losses: convex surrogates of the 0–1 loss.

Which to use?

- (Nearly) separable classes: SVM (and LDA) beat logistic regression.
- Otherwise ridge-penalized logistic regression and SVM are very similar.
- For probabilities, logistic regression.
- Nonlinear boundaries: kernel SVMs (kernels also work with logistic regression and LDA, at higher cost).

Summary

- Maximal margin: $M = 1/\|\beta\|$, determined by support vectors; soft margin with budget C.
- Kernels give nonlinear SVMs through inner products; OVA/OVO for $K > 2$.
- SVM = hinge loss + ridge penalty, close to penalized logistic regression.

7 Selected exercises

Motivation. ISLR Ex. 9.3 by hand, and validity of the radial kernel.

Exercise 12.1 (A maximal margin classifier by hand; ISLR Ex. 9.3)

Seven points in $\mathbb{R}^2$: Red $(3,4),(2,2),(4,4),(1,4)$ (observations 1–4) and Blue $(2,1),(4,3),(4,1)$ (observations 5–7). Code Red as $y=+1$, Blue as $y=-1$.
(a) Find the optimal separating hyperplane and prove that it is optimal.
(b) State the classification rule, the margin and the support vectors.
(c) Show that a slight movement of observation 7 does not change the hyperplane.

Solution. We solve $\min \frac{1}{2}\|\beta\|^2$ subject to $y_i(\beta_0+\beta_1 x_{i1}+\beta_2 x_{i2})\geq 1$.

(a)

Step 1 (a lower bound). The constraints of observations 2, 5 and 6 read

$$\beta_0+2\beta_1+2\beta_2\geq 1, \qquad -\beta_0-2\beta_1-\beta_2\geq 1, \qquad -\beta_0-4\beta_1-3\beta_2\geq 1.$$

Adding the first two gives $\beta_2\geq 2$; adding the first and third gives $-2\beta_1-\beta_2\geq 2$, so $\beta_1\leq -1-\beta_2/2\leq -2$. Hence every feasible β has $\|\beta\|^2=\beta_1^2+\beta_2^2\geq 8$, with equality only for $\beta_1=-2$, $\beta_2=2$.

Step 2 (the intercept). With $\beta_1=-2$, $\beta_2=2$, the first constraint gives $\beta_0\geq 1$ and the second $\beta_0\leq 1$, so $\beta_0=1$.

Step 3 (feasibility). $f(x)=1-2x_1+2x_2$ takes the values $3,1,1,7$ at the Red points and $-1,-1,-5$ at the Blue points, so all constraints hold. By Steps 1–2 it is the unique optimum. The hyperplane is $1-2X_1+2X_2=0$, i.e. $0.5-X_1+X_2=0$.

(b) Classify to Red if $0.5-X_1+X_2>0$ and to Blue otherwise. The margin is $M=1/\|\beta\|=1/\sqrt{8}\approx 0.354$. The support vectors are the points with $y_i f(x_i)=1$: observations 2, 3, 5 and 6.

(c) Observation 7 has $y_7 f(x_7)=5$. If it moves by δ with $\|\delta\|<\sqrt{2}$, then $|f(x_7+\delta)-f(x_7)|=|\beta^\top\delta|\leq\sqrt{8}\|\delta\|<4$, so its constraint still holds and (β_0,β) remains feasible. The lower bound of Step 1 and Step 2 used only observations 2, 5 and 6, which did not move; so the same hyperplane is still the unique optimum. ■

Exercise 12.2 (The radial kernel is positive semidefinite)

Show that for any $x_1,\dots,x_n\in\mathbb{R}^p$ and $\gamma>0$ the matrix $K=\left[\exp(-\gamma\|x_i-x_j\|^2)\right]_{i,j}$ is positive semidefinite.

Solution [Supplement].

Step 1. The Gram matrix $G=[x_i^\top x_j]$ is PSD: $c^\top Gc=\|\sum_i c_i x_i\|^2\geq 0$.

Step 2 (Schur product). If A,B are PSD, so is the entrywise product $A\circ B$. Write $B=\sum_k \lambda_k v_k v_k^\top$ with $\lambda_k\geq 0$; then

$$c^\top(A\circ B)c=\sum_k\lambda_k\sum_{i,j}(c_i v_{ki})A_{ij}(c_j v_{kj})=\sum_k\lambda_k(c\circ v_k)^\top A(c\circ v_k)\geq 0.$$

Step 3. By induction, every entrywise power $G^{\circ m}$ is PSD ($G^{\circ 0}=\mathbf{1}\mathbf{1}^\top$). Hence $E_M=\sum_{m=0}^{M}\frac{(2\gamma)^m}{m!}G^{\circ m}$ is PSD, and $E_M\to E=\left[e^{2\gamma x_i^\top x_j}\right]$ entrywise; since $c^\top E_M c\geq 0$ for every M, also $c^\top Ec\geq 0$.

Step 4. $\|x_i-x_j\|^2=\|x_i\|^2-2x_i^\top x_j+\|x_j\|^2$ gives $K=DED$ with $D=\operatorname{diag}(e^{-\gamma\|x_i\|^2})$, so $c^\top Kc=(Dc)^\top E(Dc)\geq 0$.

By Mercer, the radial kernel is an inner product in some feature space. ■